海鲸药业
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鲑降钙素鼻喷雾剂

Calcitonin (Salmon) Nasal Spray

医学知识培训

海鲸药业·中央市场医学部·医学部 王旻昊

2026-06-22

培训时长：30分钟

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- 鲑降钙素鼻喷雾剂
  - 是什么
    - 初识降钙素
      - 调控血钙水平的多肽激素
    - 降钙素的生理作用
      - 调节钙稳态
        - 降低血清钙浓度
      - 作用于破骨细胞
        - 减少骨吸收
      - 作用于肾小管
        - 抑制钙磷重吸收
      - 作用于中枢&外周神经系统
        - 改善疼痛
    - 鲑降钙素的来源
      - 鲑鱼
        - 鳃后体/终末腮体
        - 效价为人降钙素的40~50倍
          - 原因：海水高钙
    - 降钙素的药理作用
      - 降钙
      - 抑制骨吸收
      - 镇痛
    - 鲑降钙素的药代动力学
      - 吸收、分布、代谢、排泄
      - 鼻喷剂型的特点、优势、循环过程
    - 鲑降钙素的安全性：恶性肿瘤风险上升?
    - 降钙素家族：鲑降钙素vs依降钙素?
  - 为什么
    - 临床试验
      - PROOF试验
      - QUEST试验
      - 其他
    - 系统评价与Meta分析
      - JAMA Network Open
        - 降钙素对急性椎体压缩性骨折疼痛的改善作用

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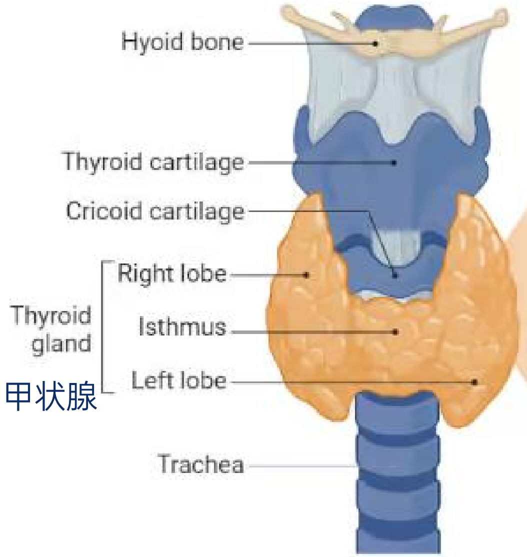
通过降钙素发挥药理作用

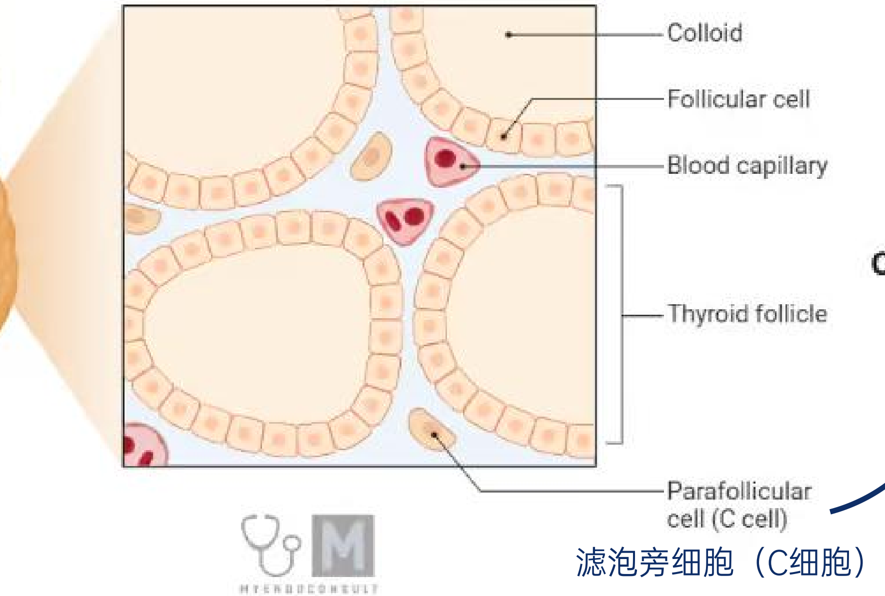
鲑降钙素鼻喷雾剂

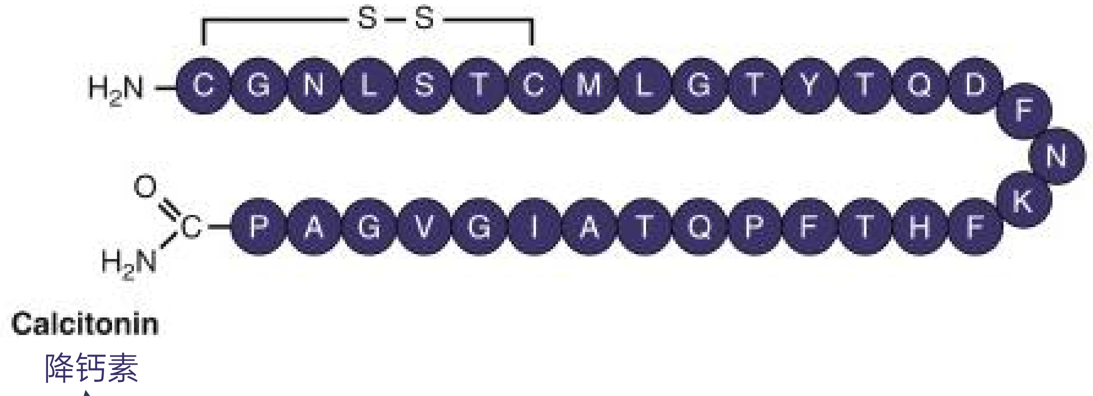
氨基酸序列来源于鲑鱼

通过鼻黏膜吸收进入体循环

降钙素是什么：调控血钙水平的多肽激素







多肽类激素
降钙素是一种多肽类激素[1,3]

由甲状腺滤泡旁细胞分泌[1]
也称C细胞[3]

通过降钙素受体发挥作用[1]
钙和胃泌素等肠道激素刺激降钙素分泌[1,4]

发挥降低血钙水平作用[3]
4~6 h即可迅速降低血钙[5]

Kiriakopoulos A, Giannakis P, Menenakos E. Calcitonin: current concepts and differential diagnosis[J]. Therapeutic advances in endocrinology and metabolism, 2022, 13: 20420188221099344.（Q1，IF=4.6）
https://www.ncbi.nlm.nih.gov/sites/books/NBK537269/
https://selfhacked.com/blog/calcitonin-miacalcin/
Cornish J, Naot D, Martin T J. Calcitonin: its physiological role and emerging therapeutics[M]//Bone-Metabolic Functions and Modulators. London: Springer London, 2012: 101-112.
https://pharmacyfreak.com/mechanism-of-action-of-calcitonin

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降钙素的生理作用

1

调节钙稳态

拮抗甲状旁腺激素（PTH）的作用，并降低血清钙浓度

2

作用于破骨细胞

抑制破骨细胞，减少骨吸收：它诱导破骨细胞运动静止，伪足回缩，膜皱褶停止

3

作用于肾小管

降低钙磷：抑制近端和远端肾小管对磷酸盐和钙的重吸收，促进磷酸盐和钙的尿排泄

4

作用于中枢&外周神经系统

改善疼痛：偏头痛患者长期服用降钙素与β-内啡肽、促肾上腺皮质激素和皮质醇水平升高有关

Kiriakopoulos A, Giannakis P, Menenakos E. Calcitonin: current concepts and differential diagnosis[J]. Therapeutic advances in endocrinology and metabolism, 2022, 13: 20420188221099344.（Q1，IF=4.6）

鲑：鲑科鱼类·太平洋鲑属（Oncorhynchus[1]）

Niall H D, Keutmann H T, Copp D H, et al. Amino acid sequence of salmon ultimobranchial calcitonin[J]. Proceedings of the National Academy of Sciences, 1969, 64(2): 771-778.（Q1，IF=9.1）

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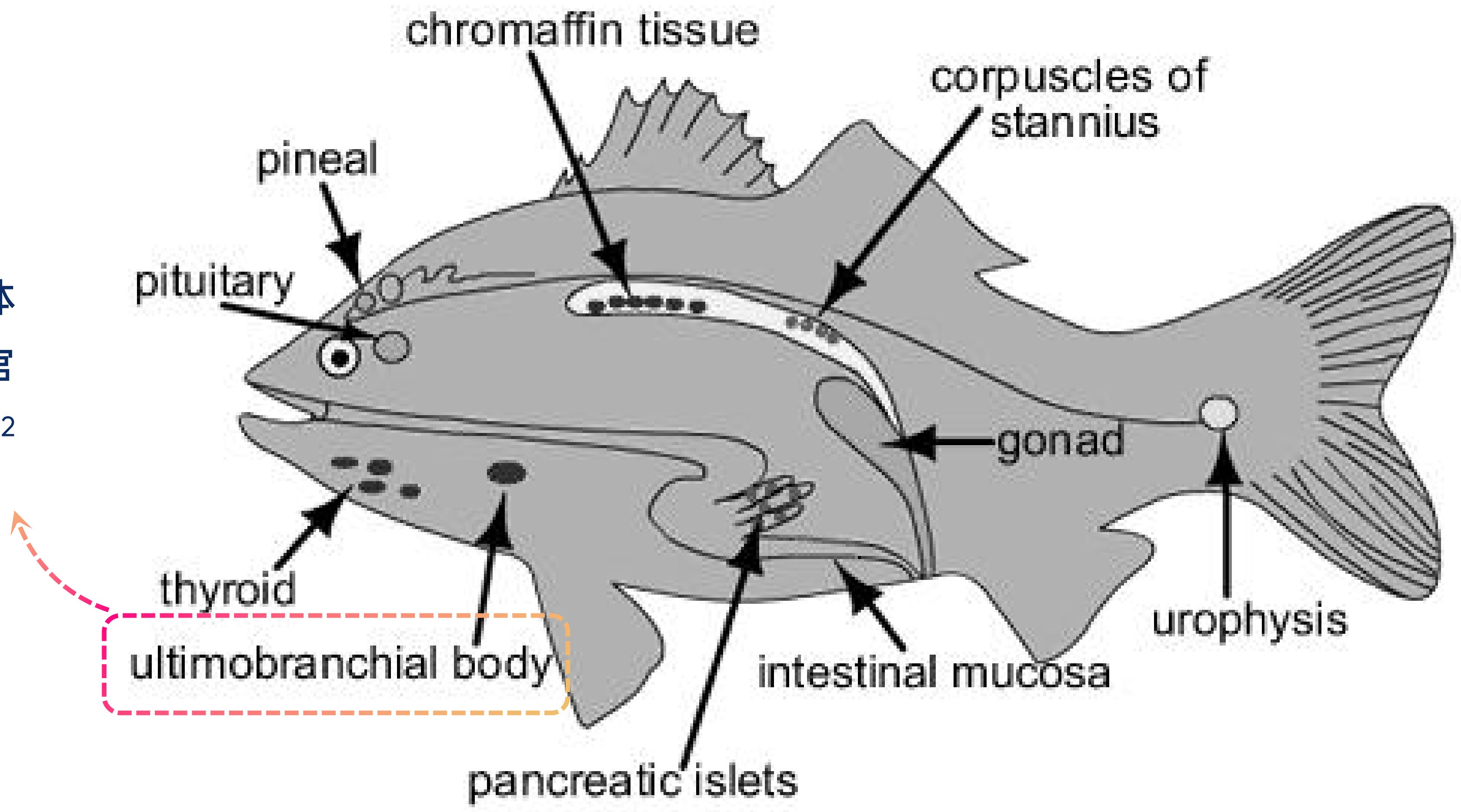
鲑降钙素：鳃后体（ultimobranchial body[1]）分泌

又称终末腮体

是硬骨鱼体内的内分泌器官

鱼类没有独立的甲状旁腺器官[2]

*但某些鱼类可分泌甲状旁腺激素[2]



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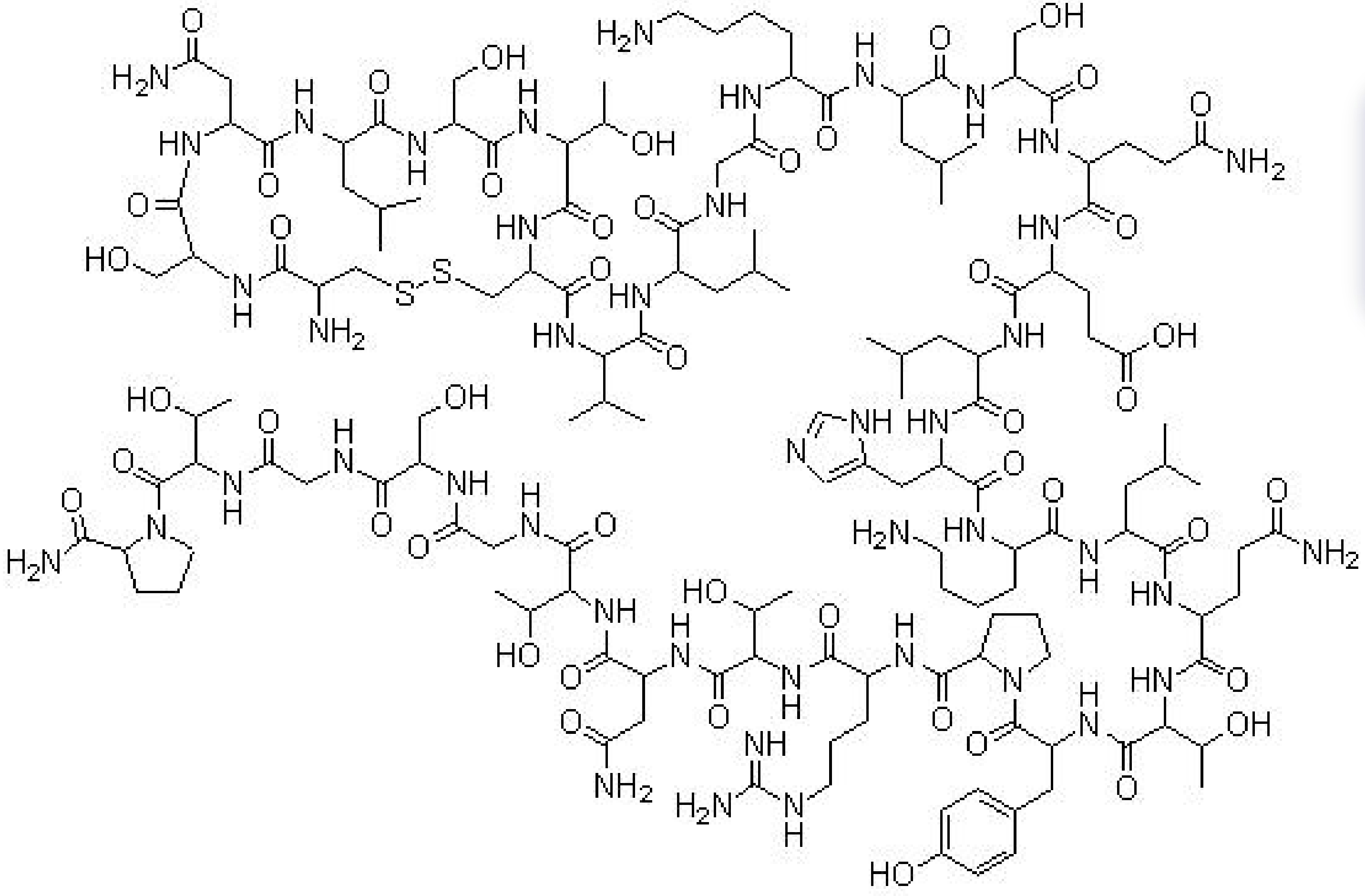
https://www.necropsymanual.net/en/teleosts-anatomy/endocrine-system

目前鲑降钙素来源于人工合成



降钙素

降钙素，又称甲状腺降钙素

是一种由32个氨基酸组成的肽类激素

分子式	$C_{145}H_{240}N_{44}O_{48}S_2$
分子量	3431.9 Da

目前大多数可通过鼻内给药进入全身循环的药物（肽类或蛋白质）的分子量在1000至3400之间[2]

https://www.chemicalbook.com/ProductChemicalPropertiesCB2127316_EN.htm

Alparslan A L, TÜRKYILMAZ G Y, Kozaci L D, et al. Thermoreversible gel formulation for the intranasal delivery of salmon calcitonin and comparison studies of in vivo bioavailability[J]. Turkish Journal of Pharmaceutical Sciences, 2023, 20(3): 127.（Q3，IF=2.0）

为什么选择鲑降钙素？其效价可达人降钙素的50倍[1,2]且较猪降钙素更高[3]

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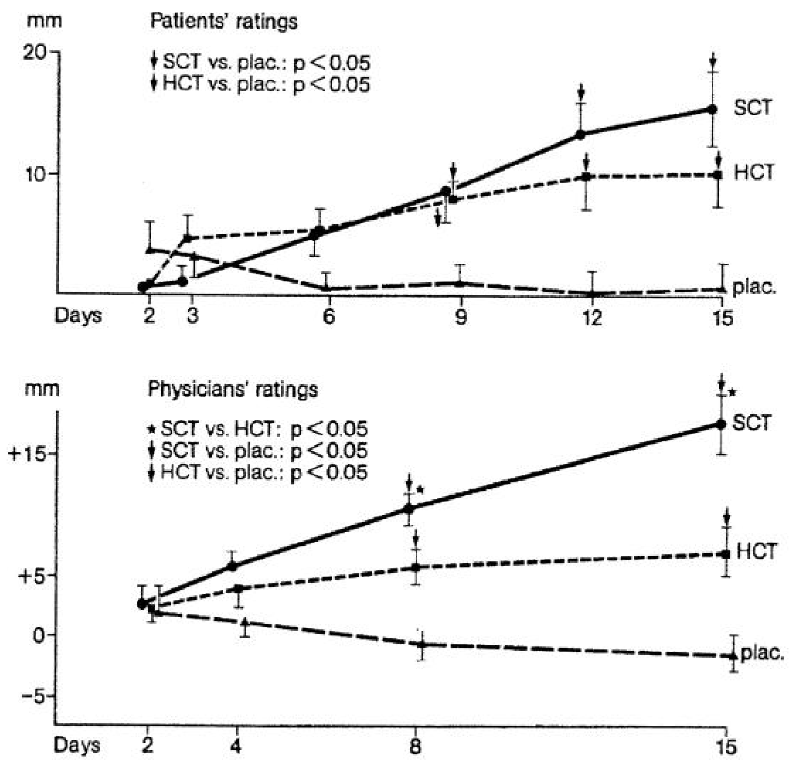
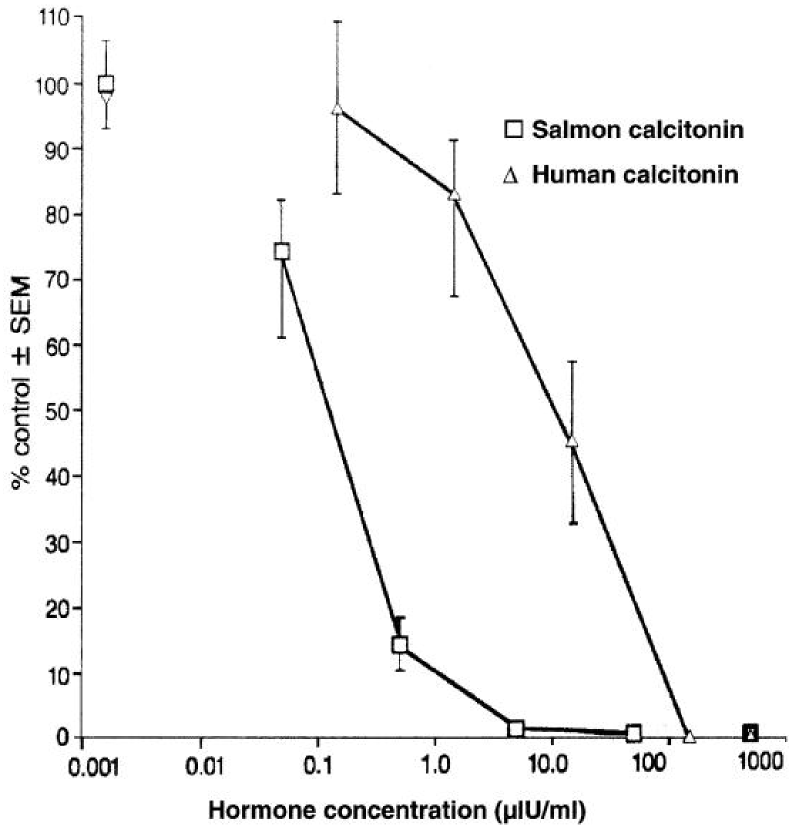
效价：每单位降钙素发挥的作用



鲑降钙素对比人降钙素：镇痛效果

二者均优于安慰剂，鲑降钙素镇痛效果更强且作用时间更长



鲑降钙素对比人降钙素：破骨细胞活性抑制

鲑降钙素对破骨细胞活性抑制作用更强

40~50倍

鲑降钙素的效价

约为人降钙素的40~50倍[1,2]

且高于猪等其他哺乳动物[3]

镇痛效果更好[4]

随用药时间延长，鲑降钙素对疼痛VAS评分的改善程度愈发显著

作用时间更长[4]

鲑降钙素作用持续时间更长

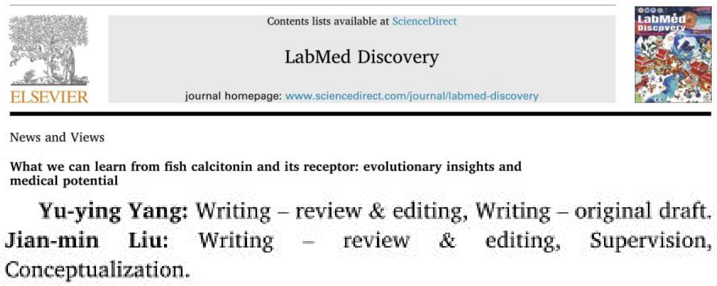
https://www.chemicalbook.com/ProductChemicalPropertiesCB2127316_EN.htm

Wells G, Chernoff J, Gilligan J P, et al. Does salmon calcitonin cause cancer? A review and meta-analysis[J]. Osteoporosis International, 2016, 27(1): 13-19.（Q1，IF=5.4）

Nafeez Ahmed A, Abdul Majeed S, Taju G, et al. Production of biologically active recombinant salmon calcitonin in Escherichia coli and fish cell line[J]. Archives of Microbiology, 2025, 207(2): 44.（Q3，IF=2.6）

Azria, M. "Possible mechanisms of the analgesic action of calcitonin." Bone 30.5 (2002): 80-83.（Q2，IF=3.6）

为什么鲑降钙素效力更高？海水环境高钙（10~10.5 mmol/L）


Contents lists available at ScienceDirect

LabMed Discovery

journal homepage: www.sciencedirect.com/journal/labmed-discovery

News and Views

What we can learn from fish calcitonin and its receptor: evolutionary insights and medical potential

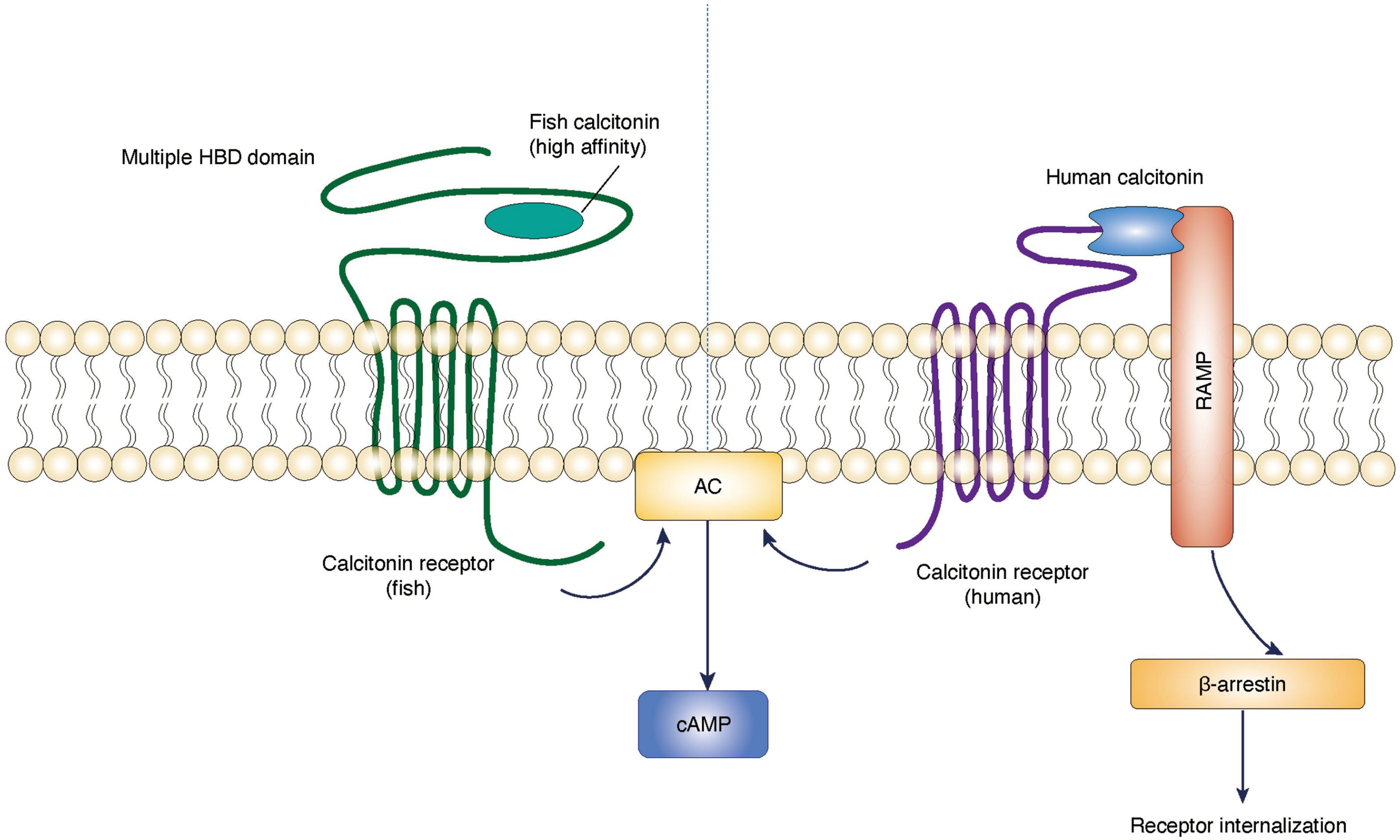
Yu-ying Yang: Writing – review & editing, Writing – original draft. **Jian-min Liu:** Writing – review & editing, Supervision, Conceptualization.

上海瑞金医院内分泌科

刘建民主任、杨毓莹医师：

在鱼类中，降钙素受体（CTR）的多个氢键结合域（HBD）和降钙素稳定的α螺旋结构共同作用，使其具有高亲和力和缓慢解离，从而触发持续的AC/cAMP信号通路。在人类中，RAMP依赖的β-arrestin募集加速了CTR的内吞作用，尽管初始AC激活动力学相似，但会破坏AC/cAMP信号通路[1]。

海洋硬骨鱼进化出一种独特的降钙素-降钙素受体（CT-CTR）系统，该系统能够抵抗脱敏，使其能够在高钙海水环境（10~10.5 mmol/L）中终生调节钙水平，从而避免致命的高钙血症[2]。



Yang Y, Liu J. What we can learn from fish calcitonin and its receptor: evolutionary insights and medical potential[J]. LabMed Discovery, 2025, 2(4): 48-50.
What we can learn from fish calcitonin and its receptor: Evolutionary insights and medical potential. https://www.eurekalert.org/news-releases/1111094

鲑降钙素的药理作用：抑制骨吸收+镇痛+降钙

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镇痛作用持续时间长于其降钙作用[1,2]

用药期间需持续补钙

Azria, M. "Possible mechanisms of the analgesic action of calcitonin." Bone 30.5 (2002): 80-83.（Q2，IF=3.6）

Toh, Sang H., C. Claunch, and Paul H. Brown. "Effect of calcitonin treatment on the natural course of bone demineralization in Paget's disease." The Journal of Clinical Endocrinology & Metabolism 56.2 (1983): 405-409.（Q1，IF=5.4）

鲑降钙素的降钙作用机制：减少钙重吸收、促进钙排出

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1

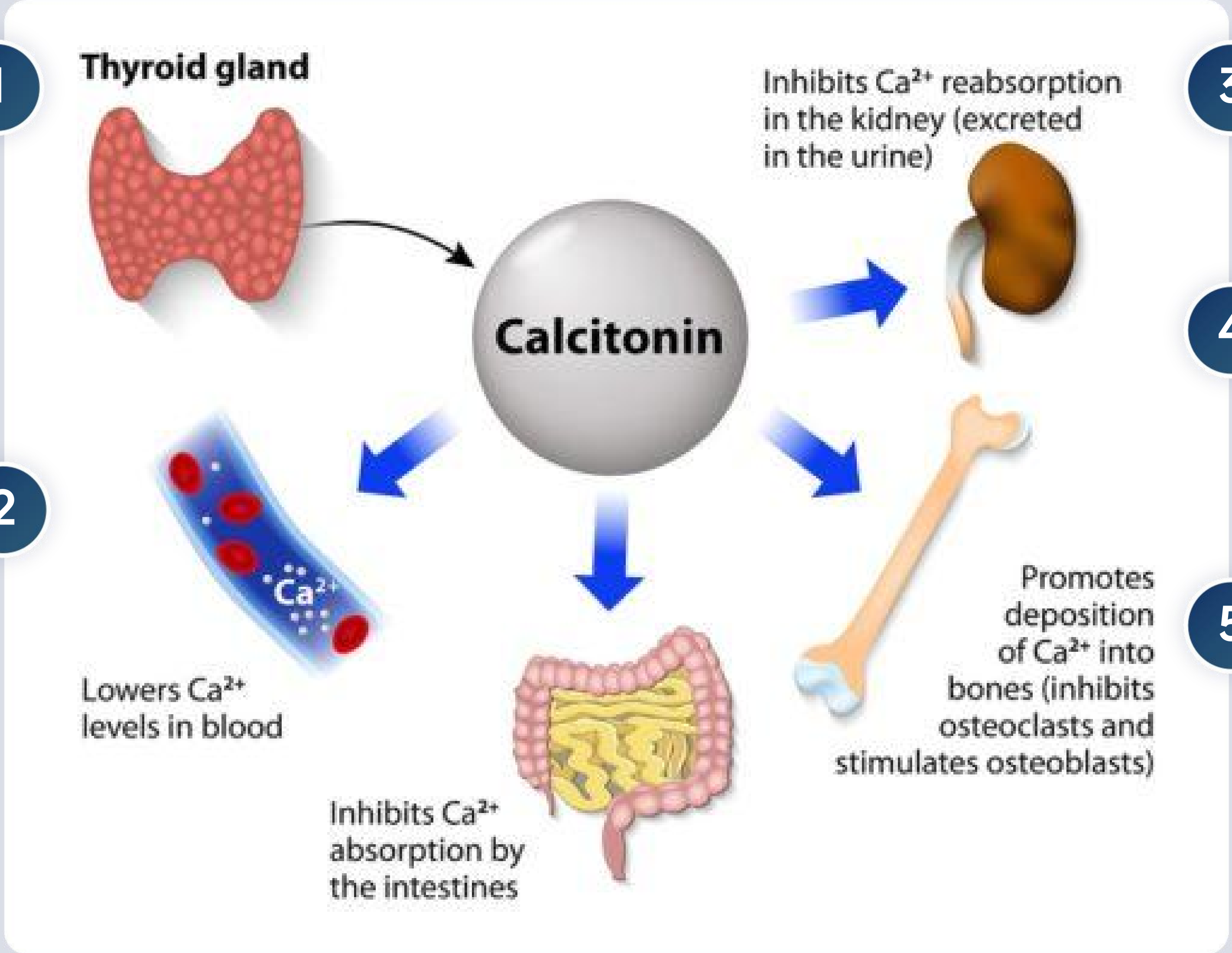
降低破骨细胞活性减少骨质流失[1]

破骨细胞活动度下降、骨吸收减少、**骨骼钙释放减少**[2]→血清钙含量降低，骨转换速度减慢

2

减少肾脏对钙的再吸收，增加尿液排出钙[1]

减少肾小管对 Ca^{2+}的重吸收、减少磷酸盐（PO_4^{3-}）的重吸收、减少钠离子（Na^+）的重吸收→降低血清钙水平[2]



3

抑制骨吸收[2]

骨转换率下降，缓解急性骨痛

4

抑制破骨细胞介导的质子泵作用[2]

阻断质子泵激活

5

调节胃肠道钙处理[2]

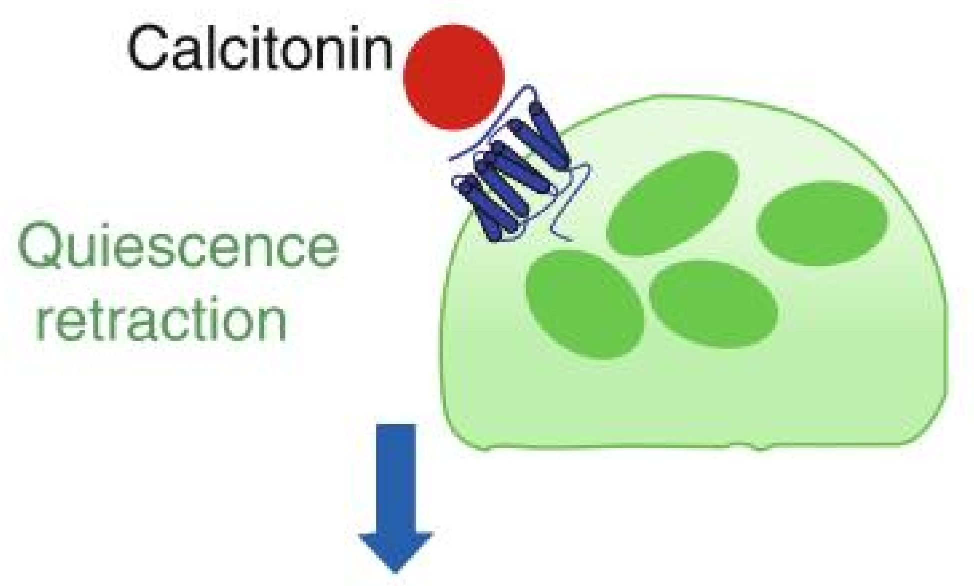
轻微降低肠道钙吸收，略微降低血清钙水平

https://selfhacked.com/blog/calcitonin-miacalcin/
https://pharmacyfreak.com/mechanism-of-action-of-calcitonin

鲑降钙素抑制骨吸收机制：结合破骨细胞

骨骼是降钙素在人体内的主要作用靶点[1]：

降钙素与破骨细胞结合，通过诱导破骨细胞进入静止状态并使其伪足回缩，从而抑制骨吸收[2]



抑制骨吸收

削弱硬骨抑素对骨形成的抑制作用

Toh S H, Claunch C, Brown P H. Effect of calcitonin treatment on the natural course of bone demineralization in Paget's disease[J]. The Journal of Clinical Endocrinology & Metabolism, 1983, 56(2): 405-409.（Q1，IF=5.4）
Cornish J, Naot D, Martin T J. Calcitonin: its physiological role and emerging therapeutics[M]//Bone-Metabolic Functions and Modulators. London: Springer London, 2012: 101-112.
邢婷婷,李燕,文罗娜,等.硬骨抑素在代谢性疾病中的研究进展[J].中华内分泌代谢杂志, 2024, 40(2):168-172.

鲑降钙素镇痛作用机制：中枢+外周双重作用机制

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骨局部机制
减少骨吸收相关疼痛刺激[1]

- 抑制破骨细胞
- 缓解机械应力、减少骨源性疼痛信号
- 减少骨吸收
- 减轻骨质疏松症或神经性疼痛患者的腰痛
- 降低局部酸性微环境刺激

中枢神经机制
下行疼痛抑制与内源性阿片样通路[2]

- 通过中枢神经系统受体及下丘脑-垂体相关通路，促进β-内啡肽、ACTH和皮质醇释放，增强下行疼痛抑制并减轻炎症相关疼痛刺激[2]
- 调节神经组织与脑脊液之间离子钙转运的变化[3]，提高疼痛阈值
- 抑制周围神经疼痛刺激引起的神经元行为和放电[3]

炎症介质机制
前列腺素/血栓素相关调节[3]

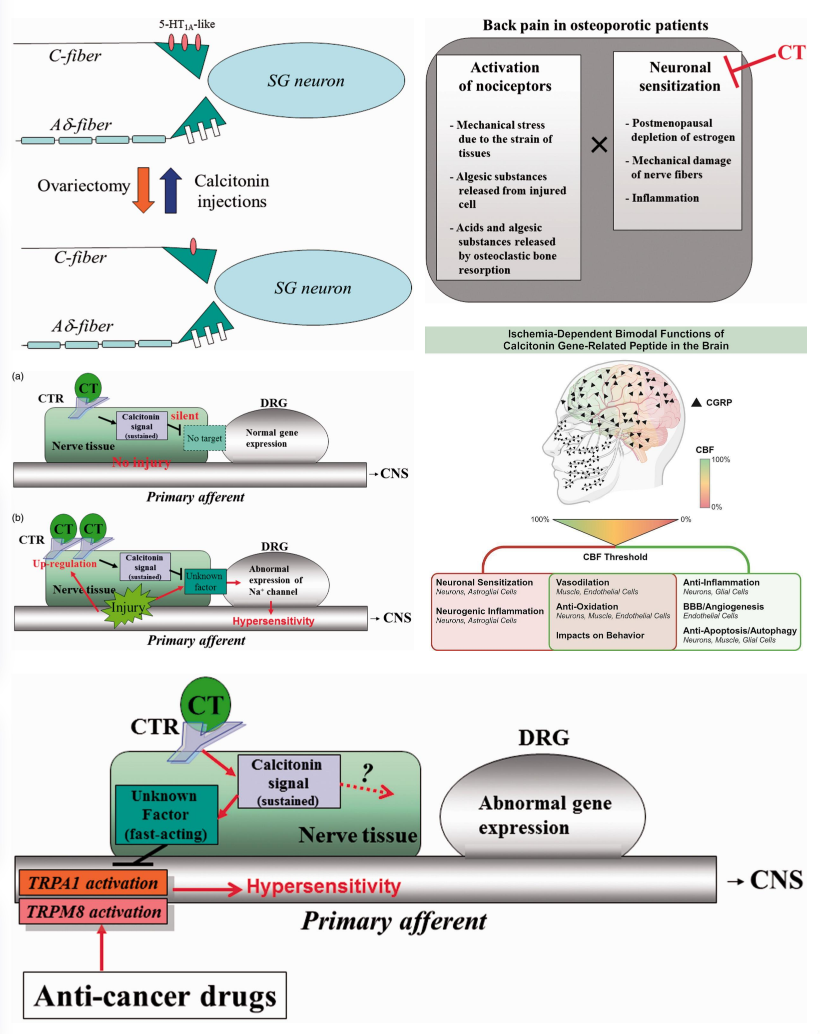
- 影响前列腺素、血栓素等炎症与血管活性介质，降低外周致痛刺激和神经敏化
- 减少吗啡类镇痛药需求（钙离子能拮抗吗啡引起的镇痛作用）

神经可塑性机制
5-HT受体、Na^+通道等[3]

- 降钙素可能通过恢复5-HT受体表达、正常化Na^+通道表达，并抑制TRPA1/TRPM8相关信号，降低初级传入神经兴奋性和痛觉过敏

CGRP相关机制[4]

- 降钙素基因相关肽（CGRP）已被证实与血管扩张、神经源性炎症和神经元敏化有关，鲑降钙素通过CGRP通路发挥疼痛调控作用



Yazdani J, Khiavi R K, Ghavimi M A, et al. Calcitonin as an analgesic agent: review of mechanisms of action and clinical applications[J]. Brazilian Journal of Anesthesiology (English Edition), 2019, 69(6): 594-604.（Q2，IF=1.9）
Ito, Akitoshi, and Megumu Yoshimura. "Mechanisms of the analgesic effect of calcitonin on chronic pain by alteration of receptor or channel expression." Molecular Pain 13 (2017): 1744806917720316.（Q2，IF=2.8）
Azria, M. "Possible mechanisms of the analgesic action of calcitonin." Bone 30.5 (2002): 80-83.（Q2，IF=3.6）
Lin K, Stiles J, Tambo W, et al. Bimodal functions of calcitonin gene-related peptide in the brain[J]. Life Sciences, 2024, 359: 123177.（Q1，IF=6.4）

鲑降钙素镇痛作用机制汇总：中枢提高疼痛阈值+外周减轻疼痛刺激

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中枢机制

- 通过中枢受体、内源性阿片样通路和下行疼痛调控，提高疼痛阈值、减轻痛觉敏化

外周机制

- 抑制破骨细胞、减少骨吸收，降低骨局部酸性微环境
- 抑制炎症介质和机械刺激相关疼痛信号

药代动力学

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吸收

进入体循环低

鼻黏膜吸收存在上限

- 降钙素的生物利用度因给药途径而异（鼻喷剂、皮下注射剂和肌注剂）[1]。
- **鼻内给药**可通过鼻黏膜迅速吸收，给药1 h内即可达到血浆峰浓度（中位时间约10 min），**生物利用度为3%~5%**[3]。
- **皮下注射和肌注**可在15 ~30 min内达到血浆峰浓度，生物利用度约为66%。高于推荐剂量的给药将会使血药浓度升高，但相对生物利用度并不随之增加[3]。

分布

血浆清除快 无蓄积效应

- 降钙素吸收后会扩散至全身。该药物的半衰期相对较短，仅为几分钟至几小时[1]。
- 降钙素易与血浆蛋白（主要是白蛋白）结合，结合率约为30%~40%。该药物分布于各种身体组织，其中**肾脏、骨骼和中枢神经系统**的浓度最高。此外，降钙素的分布容积为0.15~0.3L/kg[1]。

代谢

主要通过肾脏代谢

- 降钙素的主要代谢过程涉及蛋白水解酶的酶促分解，主要发生在肾脏和其他组织中[1]。
- 主要在肾脏代谢，血浆半衰期约5 min[2] 。

排泄

主要通过肾脏排出

- 95%的鲑降钙素及其代谢物经肾脏排出，其中2%为原型[3]（其余为：未明确鉴定的、无活性代谢物[4]）。降钙素及其代谢物通过尿液和粪便排出体外[1]。

注意事项[3]：多肽类激素血药浓度并非治疗效应的直接预测因子，故应关注临床疗效参数而非药代动力学参数

https://www.ncbi.nlm.nih.gov/sites/books/NBK537269/
Cornish J, Naot D, Martin T J. Calcitonin: its physiological role and emerging therapeutics[M]//Bone-Metabolic Functions and Modulators. London: Springer London, 2012: 101-112.
鲑降钙素鼻喷雾剂说明书（达芬盖®，2023-01-01）；鲑降钙素鼻喷雾剂说明书（密盖息®，2024-03-15）
MIACALCIC®1(salcatonin)说明书（2015-09-08）

药代动力学：鼻喷剂型的特点与优势

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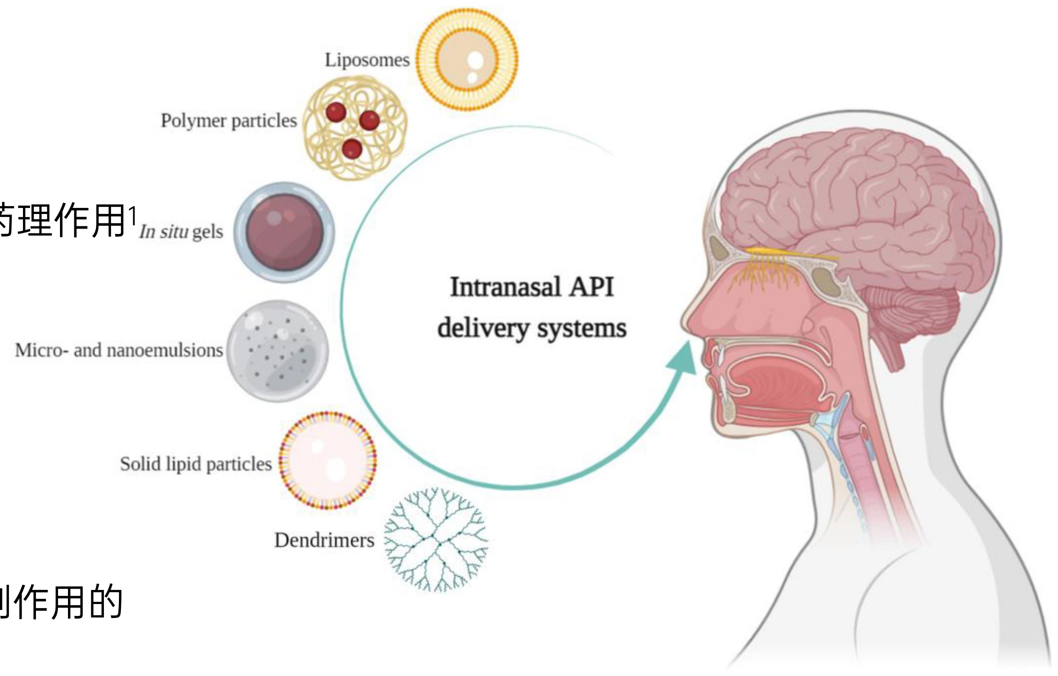
1 鼻腔血管丰富[1]

- 鼻腔血管丰富，且鼻内给药途径可以避免肝脏的首过代谢，快速发挥药理作用[1]
- 由于鼻腔易于操作，鼻内给药途径的优势在于患者可以自行用药[1]
- 大多数用于治疗鼻黏膜局部炎症的鼻内药物是小分子药物

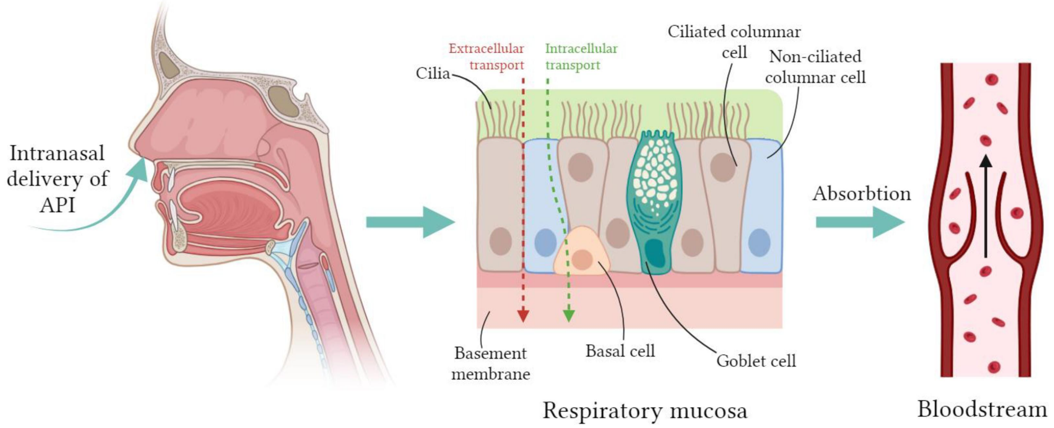


2 降低全身副作用风险[2]

- 与口服给药相比，鼻腔制剂所需的活性成分剂量更低。降低了全身性副作用的风险[2]

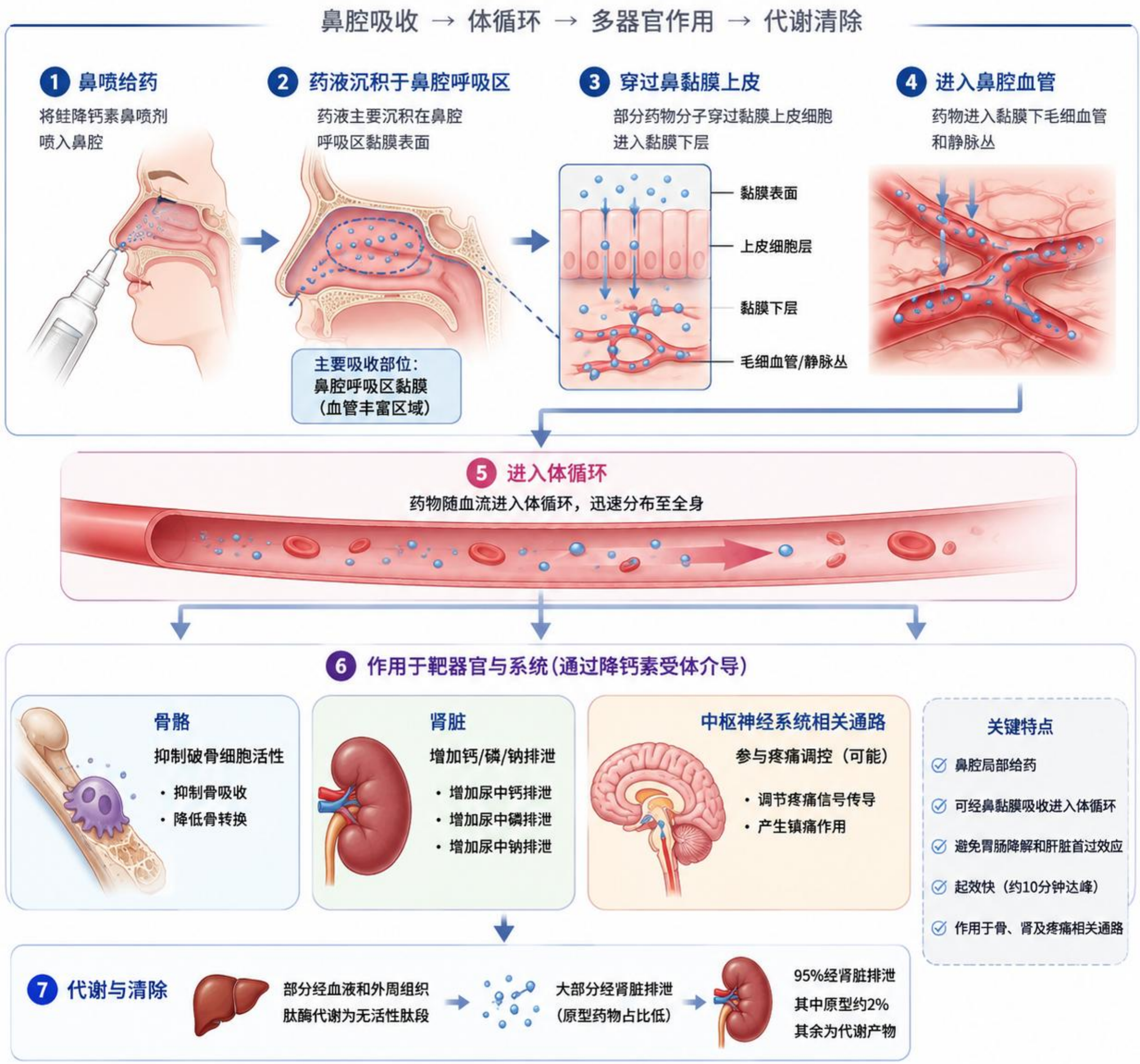
3 非侵入性、便利给药

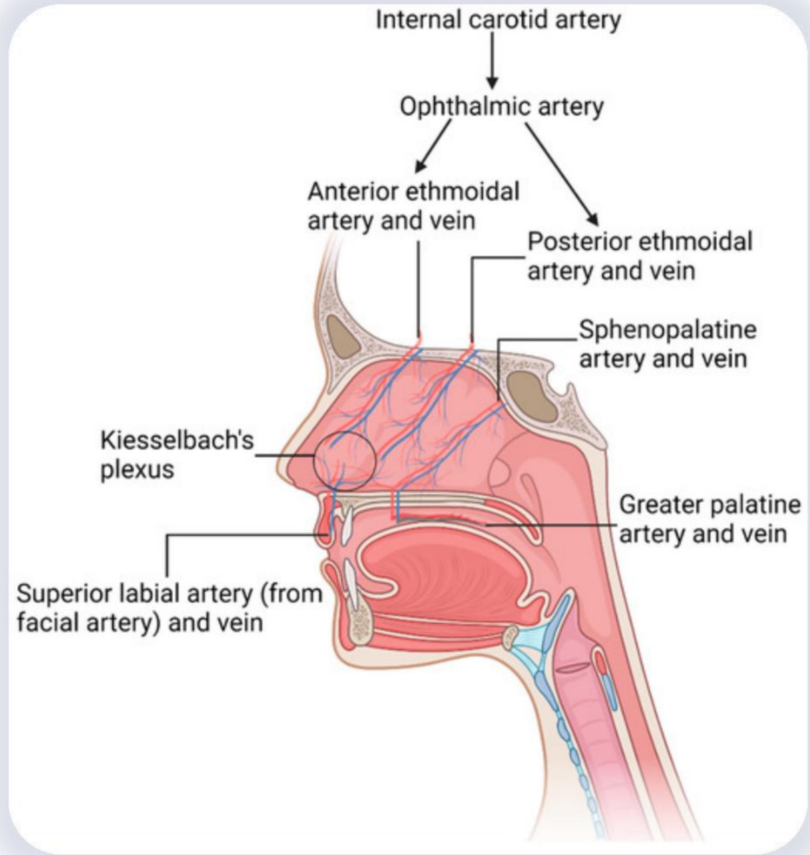
- 活性药物成分（API）从鼻上皮直接进入血液循环主要通过细胞内和/或细胞外途径[2]



Maeng J, Lee K. Systemic and brain delivery of antidiabetic peptides through nasal administration using cell-penetrating peptides[J]. Frontiers in pharmacology, 2022, 13: 1068495.（Q1，IF=4.8）
Safarov R, Fedotova O, Uvarova A, et al. Review of intranasal active pharmaceutical ingredient delivery systems[J]. Pharmaceuticals, 2024, 17(9): 1180.（Q1，IF=5.7）

药代动力学：鼻喷剂型后体内循环过程





鼻黏膜中丰富的静脉

有利于药物的有效全身靶向给药[1]

Usman Khan M, Cai X, Shen Z, et al. Challenges in the development and application of organ-on-chips for intranasal drug delivery studies[J]. Pharmaceutics, 2023, 15(5): 1557.（Q1，IF=6.9）

鼻喷400 IU与注射30 IU生物学效应相当[1]，不适、副作用更少[2]且不影响内源性CT

海鲸药业
Haijing Pharmaceutical

Clinical Science (1990) 78, 215–219 215

A comparison of the acute effects of subcutaneous and intranasal calcitonin

D. P. O'DOHERTY, D. R. BICKERSTAFF, E. V. McCLOSKEY, R. ATKINS, N. A. T. HAMDY AND J. A. KANIS
Department of Human Metabolism and Clinical Biochemistry, University of Sheffield Medical School, Sheffield, U.K.

4. Four hundred units of calcitonin administered as a nasal spray induced effects qualitatively similar to those seen with subcutaneous calcitonin, with an efficacy equivalent to approximately 30 units of subcutaneous calcitonin.

鼻内给药400 IU的降钙素与皮下注射30 IU的降钙素**具有生物等效性**[1]。较低剂量的降钙素对骨转换率较低的患者有显著疗效[1]。

ARTICLE (SCIENTIFIC JOURNALS)

Assessment of the Biological Effectiveness of Nasal Synthetic Salmon Calcitonin (Ssct) by Comparison with Intramuscular (I.M.) or Placebo Injection in Normal Subjects

Reginster, Jean-Yves; Denis, D.; Albert, Adelin *et al.*
1987 • In *Bone and Mineral, 2* (2), p. 133-40

Abstract : [en] For patients who require treatment over a period of some years, intranasal administration of synthetic salmon calcitonin (SSCT) obviates the discomfort associated with administration by injection. Moreover, this mode of administration is not associated with the side effects normally encountered when calcitonin is injected intramuscularly or subcutaneously. The aim of this study was to assess, in normal subjects, the biological activity of nasal SSCT by comparing the fluctuations of parameters reflecting calcium-phosphorus metabolism after nasal instillation, injection of SSCT and injection of placebo, respectively. In nine healthy subjects, this instillation of 200 IU of SSCT into the nasal cavity caused a fall in serum calcium, a fall in serum phosphorus and a transient rise in parathyroid hormone levels similar to that observed after the intramuscular (i.m.) injection of 80 IU of SSCT. SSCT whether administered by the nasal route or by injection, does not inhibit endogenous calcitonin secretion. There were no changes in serum beta-endorphin, magnesium or erythrocyte magnesium levels after administration of calcitonin by the intranasal route or by injection.

- 鼻内给药合成鲑鱼降钙素可**避免注射给药带来的不适**。此外，这种给药方式**不会产生**肌内或皮下注射降钙素时**常见的副作用**[2]。
- 二者均**不抑制内源性降钙素的分泌**[2]。

KANIS J A. A comparison of the acute effects of subcutaneous and intranasal calcitonin[J]. Clinical Science, 1990, 78: 215-219.（Q1，IF=7.3）
Reginster J Y, Denis D, Albert A, et al. Assessment of the biological effectiveness of nasal synthetic salmon calcitonin (SSCT) by comparison with intramuscular (im) or placebo injection in normal subjects[J]. Bone and mineral, 1987, 2(2): 133-1

嵌套病例对照研究：乳腺癌风险下降，高剂量使用使女性肝癌风险上升

海鲸药业
Haijing Pharmaceutical

Table 2. ORs and 95% CIs for Different Cancer Types Compared With Patients Without Calcitonin Use by Gender

	All		Women		Men	
Cancer type	n	OR (95% CI)	n	OR (95% CI)	n	OR (95% CI)
All cancer	1925	0.91 (0.73–1.12)	1355	0.98 (0.77–1.25)	570	0.70 (0.44–1.12)
Lung	248	1.39 (0.90–2.14)	154	1.65 (0.99–2.71)	94	1.00 (0.42–2.37)
Liver	248	1.45 (0.95–2.22)	169	1.94 (1.23–3.05)[b]	79	0.38 (0.09–1.59)
Colorectal	278	0.96 (0.60–1.54)	212	0.95 (0.56–1.62)	66	0.95 (0.33–2.68)
Female breast	206	0.35 (0.15–0.80)[a]	206	0.35 (0.15–0.80)[a]		
Prostate	96	0.64 (0.23–1.79)			96	0.64 (0.23–1.79)
Head and neck	62	0.41 (0.10–1.70)	22	0.56 (0.07–4.16)	40	0.38 (0.05–2.78)
Other	787	0.81 (0.59–1.11)	592	0.82 (0.58–1.17)	195	0.71 (0.35–1.45)

[a] $P < .05$.
[b] $P < .01$.

与未使用鲑降钙素者相比

女性肝癌风险上升1.94倍（*P*<0.01）

乳腺癌风险下降0.35倍（*P*<0.05）

Table 4. Association Between Cancer and DDD per Month of Calcitonin in Women

	All			Without Bisphosphonates			With Bisphosphonates		
	n/N	OR	95% CI	n/N	OR	95% CI	n/N	OR	95% CI
All cancer									
None	1250/3716	1.00	Reference	1155/3451	1.00	Reference	95/265	1.00	Reference
1–14 DDD	51/168	0.86	0.62–1.20	23/93	0.65	0.41–1.05	28/75	1.07	0.63–1.81
>14 DDD	54/148	1.13	0.81–1.60	29/84	1.05	0.67–1.65	25/64	1.15	0.65–2.01
Lung cancer									
None	135/2601	1.00	Reference	122/2418	1.00	Reference	13/183	1.00	Reference
1–14 DDD	6/123	0.94	0.41–2.17	3/73	0.81	0.25–2.60	3/50	0.84	0.23–3.05
>14 DDD	13/107	2.53	1.38–4.63[b]	6/61	2.05	0.87–4.86	7/46	2.35	0.88–6.27
Liver cancer									
None	145/2611	1.00	Reference	134/2430	1.00	Reference	11/181	1.00	Reference
1–14 DDD	10/127	1.45	0.75–2.83	4/74	0.98	0.35–2.72	6/53	1.97	0.69–5.62
>14 DDD	14/108	2.53	1.41–4.55[b]	8/63	2.49	1.16–5.34[a]	6/45	2.38	0.83–6.82
Female breast									
None	200/2666	1.00	Reference	192/2488	1.00	Reference	8/178	1.00	Reference
1–14 DDD	3/120	0.32	0.10–1.00	2/72	0.34	0.08–1.41	1/48	0.45	0.06–3.71
>14 DDD	3/97	0.39	0.12–1.25	3/58	0.65	0.20–2.10	0/39		

每月使用鲑降钙素达到日剂量上限>14 d

且未联合双膦酸盐

与肝癌风险上升有关（*P*<0.05）

*作者强调：肝癌病例数为6例，统计效能不足

Sun L M, Lin M C, Muo C H, et al. Calcitonin nasal spray and increased cancer risk: a population-based nested case-control study[J]. The Journal of Clinical Endocrinology & Metabolism, 2014, 99(11): 4259-4264.（Q1，IF=5.4）

致癌性观念的演变：相关性而非因果性

海鲸药业
Haijing Pharmaceutical

Osteoporos Int
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REVIEW

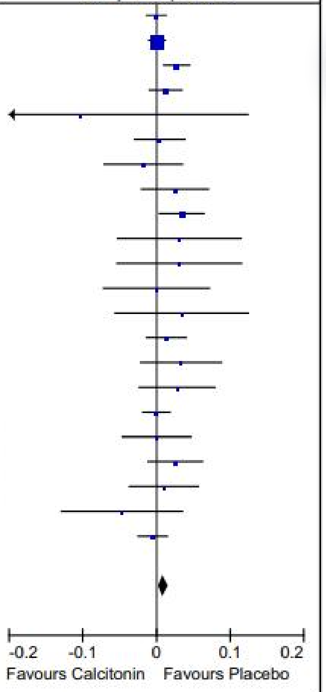
Does salmon calcitonin cause cancer? A review and meta-analysis

G. Wells[1] · J. Chernoff[2] · J. P. Gilligan[3] · D. S. Krause[3]

Study or Subgroup	Calcitonin Events	Calcitonin Total	Placebo Events	Placebo Total	Weight	Risk Difference M-H, Fixed, 95% CI
2402	0	149	0	147	2.8%	0.0000 [-0.0131, 0.0131]
A2303	89	2334	87	2331	43.9%	0.0008 [-0.0101, 0.0117]
C2301	22	585	6	584	11.0%	0.0273 [0.0099, 0.0448]
C2302	21	520	14	508	9.7%	0.0128 [-0.0093, 0.0349]
CT211	3	31	3	15	0.4%	-0.1032 [-0.3308, 0.1244]
CT310	4	211	1	68	1.9%	0.0043 [-0.0298, 0.0383]
CT311	8	244	4	79	2.2%	-0.0178 [-0.0711, 0.0354]
CT312	11	201	3	102	2.5%	0.0253 [-0.0201, 0.0707]
CT320	81	944	16	311	8.8%	0.0344 [0.0040, 0.0647]
MIA 16	1	32	0	30	0.6%	0.0313 [-0.0525, 0.1150]
SMCO 005	1	32	0	29	0.6%	0.0313 [-0.0533, 0.1158]
SMCO 503	0	26	0	26	0.5%	0.0000 [-0.0719, 0.0719]
SMCO 504	1	29	0	29	0.5%	0.0345 [-0.0560, 0.1249]
SMCO 506	3	147	1	141	2.7%	0.0133 [-0.0134, 0.0400]
SMCO 511	2	60	0	60	1.1%	0.0333 [-0.0213, 0.0880]
SMCO 514	2	71	0	46	1.1%	0.0282 [-0.0233, 0.0796]
SMCO 517	0	168	0	83	2.1%	0.0000 [-0.0184, 0.0184]
SMCO 520	0	65	0	32	0.8%	0.0000 [-0.0466, 0.0466]
SMCO 522	4	156	0	52	1.5%	0.0256 [-0.0112, 0.0625]
SMCO 524	1	100	0	33	0.9%	0.0100 [-0.0368, 0.0568]
Study 201	2	87	3	43	1.1%	-0.0468 [-0.1292, 0.0356]
Study 801	2	444	1	104	3.2%	-0.0051 [-0.0249, 0.0147]
Total (95% CI)		**6636**		**4853**	**100.0%**	**0.0090 [0.0021, 0.0158]**
Total events	258		139			

Heterogeneity: Chi² = 21.31, df = 21 (P = 0.44); I² = 1%
Test for overall effect: Z = 2.57 (P = 0.01)

Risk Difference
M-H, Fixed, 95% CI



1 研究结论

- 降钙素的使用与恶性肿瘤之间不太可能存在因果关系，因为缺乏生物学上的合理性。
- 大量的非临床和临床证据也不支持二者之间存在因果关系。

2 相关数据

- 2013年美国FDA荟萃分析显示，鲑鱼降钙素（sCT）发生任何癌症的风险差（RD）安慰剂组高 1%（95% CI，0.3，1.6）。排除基底细胞癌后，风险差异不显著，RD 为 0.5%（95% CI，-0.1，1.2）。
- 2016年Meta分析显示，得到荟萃分析风险差为 0.90%（95% CI，0.21% 至 1.58%），关联性检验表明该风险差具有统计学意义（*p*=0.01）。如前所述，数据受一项大型5年试验（CT320 [PROOF]）的显著影响，导致任何恶性肿瘤的风险略有增加。

Wells G, Chernoff J, Gilligan J P, et al. Does salmon calcitonin cause cancer? A review and meta-analysis[J]. Osteoporosis International, 2016, 27(1): 13-19.（Q1，IF=5.4）

证据：更高的降钙素水平并未显现出更高的恶性肿瘤发生风险

TABLE II – OCCURRENCE OF SECOND CANCERS FOLLOWING A FIRST PRIMARY THYROID CANCER, AND OCCURRENCE OF SECOND PRIMARY THYROID CANCER[1]

Other primary tumor	Risk of second primary thyroid cancer following any other cancer				Risk of a second primary cancer following thyroid cancer			
	O	O/E	95% CI	EAR[2]	O	O/E	95% CI	EAR[2]
All Sites[3]	1366	1.42[4]	1.35 – 1.50	0.38	2214	1.11[4]	1.06 – 1.15	7.64
Buccal cavity	42	1.70[4]	1.22 – 2.29	0.54	37	0.82	0.58 – 1.13	−0.29
Salivary gland	11	3.26[4]	1.63 – 5.84	1.94	13	2.78[4]	1.48 – 4.75	0.30
Pharynx	5	1.63	0.52 – 3.79	0.49	7	0.90	0.36 – 1.86	−0.03
Esophagus	6	3.17[4]	1.16 – 6.89	1.64	9	0.56	0.26 – 1.07	−0.25
Stomach	15	1.83[4]	1.02 – 3.01	0.71	43	1.21	0.88 – 1.63	0.27
Small intestine	3	1.47	0.29 – 4.29	0.40	9	1.42	0.65 – 2.70	0.09
Colon	97	1.16	0.94 – 1.42	0.14	182	1.06	0.92 – 1.23	0.39
Rectum	40	1.06	0.75 – 1.44	0.05	66	0.93	0.72 – 1.19	−0.17
Liver	6	7.07[4]	2.58 – 15.38	4.79	14	1.01	0.55 – 1.69	0.00
Gallbladder	0	—	0.00 – 4.23	−0.96	3	0.48	0.10 – 1.41	−0.12
Pancreas	12	4.48[4]	2.31 – 7.82	2.96	36	0.77	0.54 – 1.07	−0.38
Larynx	25	2.22[4]	1.44 – 3.27	0.88	10	0.61	0.29 – 1.13	−0.23
Lung and bronchus	84	2.10[4]	1.67 – 2.60	0.91	238	0.87[4]	0.76 – 0.99	−1.26
Trachea	1	10.32	0.13 – 57.43	7.84	5	14.47[4]	4.66 – 33.77	0.17
Female breast[5]	345	1.31[4]	1.18 – 1.46	0.35	530	1.21[4]	1.11 – 1.32	4.36
Cervix[5]	26	0.92	0.60 – 1.34	−0.10	21	0.68	0.42 – 1.04	−0.47
Uterine corpus[5]	62	0.84	0.64 – 1.08	−0.17	73	0.84	0.66 – 1.06	−0.64
Ovary[5]	32	1.36	0.93 – 1.92	0.41	55	0.99	0.74 – 1.28	−0.04
Prostate[5]	133	1.21[4]	1.01 – 1.43	0.15	257	1.31[4]	1.16 – 1.48	9.00
Testis[5]	12	1.98[4]	1.02 – 3.46	0.41	7	1.88	0.75 – 3.87	0.48
Scrotum[5]	3	11.58[4]	2.33 – 33.83	6.51	3	8.13[4]	1.63 – 23.75	0.39
Bladder	45	0.96	0.70 – 1.28	−0.03	62	0.84	0.65 – 1.08	−0.41
Kidney	45	2.93[4]	2.14 – 3.92	1.50	84	2.26[4]	1.80 – 2.79	1.67
Renal pelvis	3	1.51	0.30 – 4.41	0.41	8	2.11	0.91 – 4.16	0.15
Melanoma	93	2.01[4]	1.62 – 2.47	0.91	78	1.24	0.98 – 1.55	0.54
Brain and CNS	14	2.19[4]	1.20 – 3.68	0.72	37	1.54[4]	1.09 – 2.12	0.46
Thyroid	50	1.68[4]	1.25 – 2.21	0.72	50	1.68[4]	1.25 – 2.21	0.72
Adrenal gland	0	—	0.00 – 12.49	−0.48	4	4.72[4]	1.27 – 12.09	0.11
Bones and joints	5	2.48	0.80 – 5.80	0.93	6	2.50	0.91 – 5.44	0.13
Soft tissue	14	2.39[4]	1.30 – 4.00	1.00	13	1.41	0.75 – 2.42	0.14
NHL	38	1.29	0.91 – 1.77	0.24	71	0.99	0.77 – 1.25	−0.03
Hodgkin lymphoma	32	3.03[4]	2.07 – 4.28	1.43	14	1.67	0.91 – 2.81	0.20
Multiple myeloma	9	1.55	0.71 – 2.94	0.45	32	1.44	0.99 – 2.03	0.35
Leukemia	39	2.52[4]	1.79 – 3.45	1.01	62	1.39[4]	1.07 – 1.79	0.63

[1]U.S. SEER cancer registries, 1973–2000.–[2]Absolute Excess Risk = (O-E)/10,000 PY.–[3]excluding non-melanoma skin cancer.–[4]95% confidence interval does not include 1.0.–[5]For sex-specific cancers based on respective PY only.–O, observed number of cases; O/E ratio, observed over expected number of cases; CI, confidence interval.

该研究随访884例髓样甲状腺癌（MTC）患者开展为期8年的随访

- MTC可导致降钙素水平显著上升
- 且血清降钙素也是MTC诊断和治疗的重要指标

患者未表现出第二恶性肿瘤风险增加

因此，降钙素为特异性致癌物的假设不成立

Wells G, Chernoff J, Gilligan J P, et al. Does salmon calcitonin cause cancer? A review and meta-analysis[J]. Osteoporosis International, 2016, 27(1): 13-19.（Q1，IF=5.4）
Ronckers CM, McCarron P, Ron E. Thyroid carcinoma and multiple tumors in the SEER cancer registries. Int J Cancer. 2005;117:281–288.（Q1，IF=4.7）

我国指南推荐将鲑降钙素鼻喷雾剂疗程控制在3个月以内即可


中国全科医学 2023年5月 第26卷 第14期

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·指南·共识·

原发性骨质疏松症诊疗指南（2022）

中华医学会骨质疏松和骨矿盐疾病分会

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【关键词】 骨质疏松；骨质疏松性骨折；诊断；药物疗法；骨密度保护剂；诊疗指南
【中图分类号】 R 681 【文献标识码】 A DOI：10.12114/j.issn.1007-9572.2023.0121
中华医学会骨质疏松和骨矿盐疾病分会. 原发性骨质疏松症诊疗指南（2022）[J]. 中国全科医学, 2023, 26（14）：1671–1691. [www.chinagp.net]
Chinese Society of Osteoporosis and Bone Mineral Research. Guidelines for the diagnosis and treatment of primary osteoporosis（2022）[J]. Chinese General Practice, 2023, 26（14）：1671–1691.

Guidelines for the Diagnosis and Treatment of Primary Osteoporosis（2022） *Chinese Society of Osteoporosis and Bone Mineral Research*
Corresponding author: ZHANG Zhenlin, Department of Osteoporosis and Bone Disease, Shanghai JiaoTong University Affiliated Sixth People's Hospital; E-mail: zhangzl@sjtu.edu.cn
【Key words】 Osteoporosis; Osteoporotic fractures; Diagnosis; Drug therapy; Bone density conservation agents; Diagnostic and treatment guideline

1 概述

1.1 定义和分类 骨质疏松症（osteoporosis）是一种以骨量低下、骨组织微结构损坏，导致骨脆性增加，易发生骨折为特征的全身性骨病[1]。2001年美国国立卫生研究院（National Institutes of Health，NIH）将其定义为骨强度下降和骨折风险增加为特征的骨骼疾病[2]。骨质疏松症可发生于任何年龄，但多见于绝经后女性和老年男性。依据病因，骨质疏松症分为原发性和继发性两大类。原发性骨质疏松症包括绝经后骨质疏松症（Ⅰ型）、老年骨质疏松症（Ⅱ型）和特发性骨质疏松症（青少年型）。绝经后骨质疏松症一般发生在女性绝经后5~10年内；老年骨质疏松症一般指70岁以后发生的骨质疏松；特发性骨质疏松症主要发生在青少年，病因尚未明[3-4]。继发性骨质疏松症指由影响骨代谢的疾病或药物或其他明确病因导致的骨质疏松。本指南主要针对原发性骨质疏松症。

1.2 流行病学 随着我国人口老龄化加剧，骨质疏松症患病率快速攀升，已成为重要的公共健康问题。第七次全国人口普查显示：我国60岁以上人口为2.64亿（约占总人口的18.7%），65岁以上人口超过1.9亿（约占总人口的13.5%）[5]，是全球老年人口最多的国家。全国骨质疏松症流行病学调查显示：50岁以上人群骨质疏松症患病率为19.2%，其中女性为32.1%，男性为6.9%；65岁以上人群骨质疏松症患病率为32.0%，其中女性为51.6%，男性为10.7%[6-7]。根据以上流行病学资料估算，目前我国骨质疏松症患病人数约为9 000万[6]，其中女性约7 000万。

骨质疏松性骨折（或称脆性骨折）是指受到轻微创伤（相当于从站立高度或更低的高度跌倒）即发生的骨折，是骨质疏松症的严重后果。骨质疏松性骨折的常见部位包括椎体、前臂远端、髋部、肱骨近端和骨盆等[8]，其中椎体骨折最为常见。20世纪90年代北京地区基于影像学的椎体骨折流行病学调查显示，50岁以上女性椎体骨折患病率约为15.0%，且患病率随增龄而渐增，80岁以上女性椎体骨折患病率可高达36.6%[9]。2013年北京椎体骨折研究表明，北京地区绝经后妇女椎体骨折的患病率与20世纪90年代相似，椎体骨折的患病率呈稳定趋势[10-11]。近期上海社区人群椎体骨折筛查研究表明，60岁以上人群椎体骨折患病率：男女两性相当，其中男性为17.0%，女性17.3%[12]。全国随机抽样研究表明，我国40岁以上人群椎体骨折的患病率男性为10.5%，女性为9.5%[7]。上海和全国的数据均提示

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本文数字出版日期：2023-03-10

6.2.3 降钙素 降钙素（calcitonin）是一种钙调节激素，能抑制破骨细胞的生物活性、减少破骨细胞数量，减少骨量丢失并增加骨量[128-131]。降钙素的另一作用是能有效缓解骨痛[132-133]。目前应用于临床的降钙素制剂有两种：鳗鱼降钙素类似物依降钙素（elcatonin）（表14）和鲑降钙素（salmon calcitonin）（表15）。

降钙素总体安全性良好。2012年欧洲药品管理局（EMA）通过荟萃分析发现，长期使用（6个月或更长时间）鲑降钙素口服或鼻喷剂型与恶性肿瘤风险轻微增加相关，但无法肯定该药物与恶性肿瘤间的确切关系[134]。鉴于鼻喷剂型鲑降钙素具有潜在增加肿瘤风险的可能，鲑降钙素连续使用时间一般不超过3个月[135]。

中华医学会骨质疏松和骨矿盐疾病分会. 原发性骨质疏松症诊疗指南（2022）[J]. 中国全科医学, 2023, 26(14): 1671-1691.

鲑降钙素 vs 依降钙素（鳗鱼降钙素）：鲑降钙素缓解疼痛效果更优

海鲸药业
Haijing Pharmaceutical

doi:10.3969/j.issn.1009−4393.2013.34.060

鲑鱼降钙素和依降钙素缓解骨质疏松疼痛的临床对比研究

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作者单位：四川 610072 四川省医学科学院·四川省人民医院（杨玉龙）

[摘要] **目的** 对鲑鱼降钙素和依降钙素缓解重度骨质疏松疼痛的临床疗效进行对比。**方法** 选择 2009 年 6 月—2013 年 2 月，符合骨质疏松诊断标准、疼痛评分在 7 分以上患者 20 例，随机分为鲑鱼降钙素组 10 例、依降钙素组 10 例，分别给予肌注射鲑鱼降钙素，1 次/d，每次 50 U，和肌注射依降钙素，1 次/d，每次 20 U，均连续 2 周。观察治疗 2 周后患者的临床疗效并观察疼痛的变化，3 个月后随访观察患者的疼痛评分变化。**结果** 鲑鱼降钙素组显效效 8 例，有效 2 例；依降钙素组显效 4 例，有效 4 例，无效 2 例，鲑鱼降钙素组优于依降钙素组。疼痛评分：鲑鱼降钙素组治疗前评分(8.12±2.32)，治疗后评分(2.33±1.22)，依降钙素组治疗前评分(8.09±2.72)，治疗后评分(4.21±1.51)。两组治疗前后自身比较差异有统计学意义($P<0.05$)，治疗后组间疼痛评分比较，差异有统计学意义($P<0.05$)。**结论** 本研究表明在鲑鱼降钙素在缓解原发性骨质疏松疼痛的临床疗效上优于依降钙素。

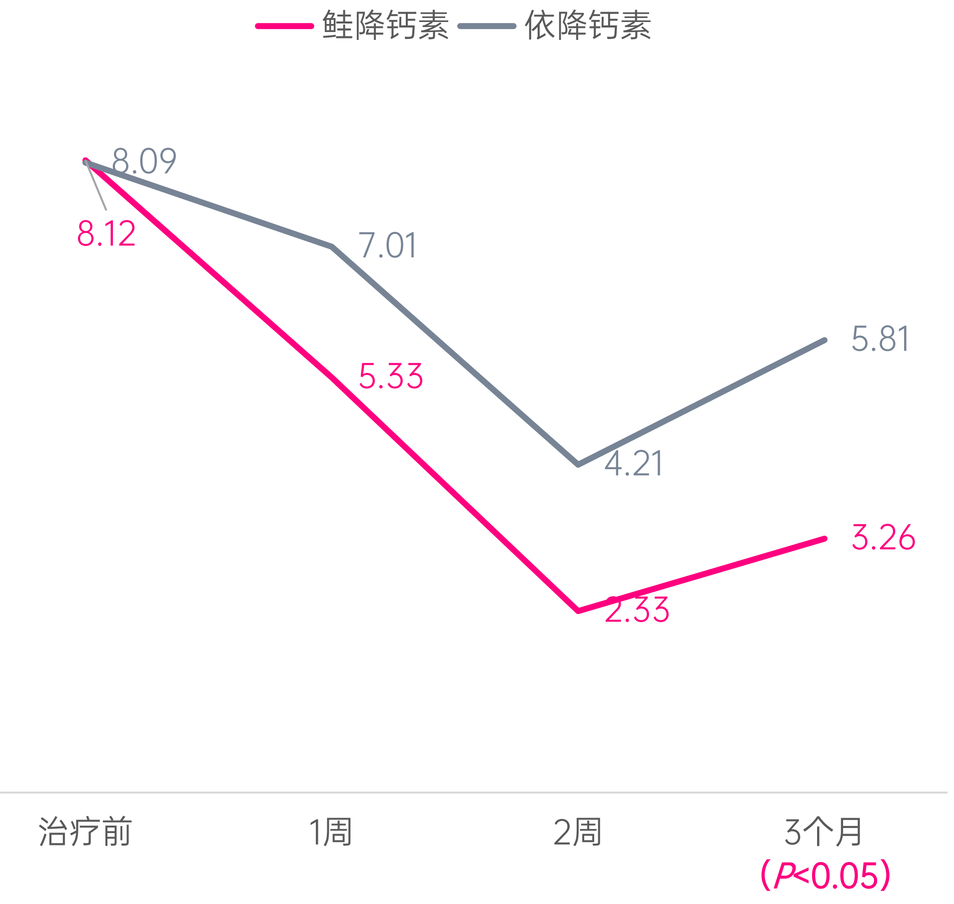
[关键词] 疼痛；骨质疏松；鲑鱼降钙素；依降钙素

表 2 两组骨质疏松患者治疗前后疼痛评分比较(分，$\bar{x}\pm s$)

组别	治疗前	1 周	2 周	3 个月
鲑鱼降钙素组	8.12±2.32	5.33±1.32	2.33±1.22	3.26±1.04[ab]
依降钙素组	8.09±2.72	7.01±2.22	4.21±1.51	5.81±1.56

注：与本组治疗前比较，[a]$P<0.05$；两组组间比较，[b]$P<0.05$

2组患者VAS评分比较（分）



杨玉龙. 鲑鱼降钙素和依降钙素缓解骨质疏松疼痛的临床对比研究[J].当代医学, 2013, 34: 78-79.

PROOF试验：200 IU鲑降钙素有效预防骨质疏松性骨折复发

PROOF试验 5年+双盲+随机+安慰剂对照

研究设计 总计1255例 骨质疏松症绝经后骨折女性

- 安慰剂组
- 100 IU 组
- 200 IU 组
- 400 IU 组

所有参与者每日均补充元素钙（1000 mg）和维生素D（400 IU）

主要终点 比较200 IU鲑降钙素鼻喷雾剂与安慰剂组的新发椎体骨折发生率

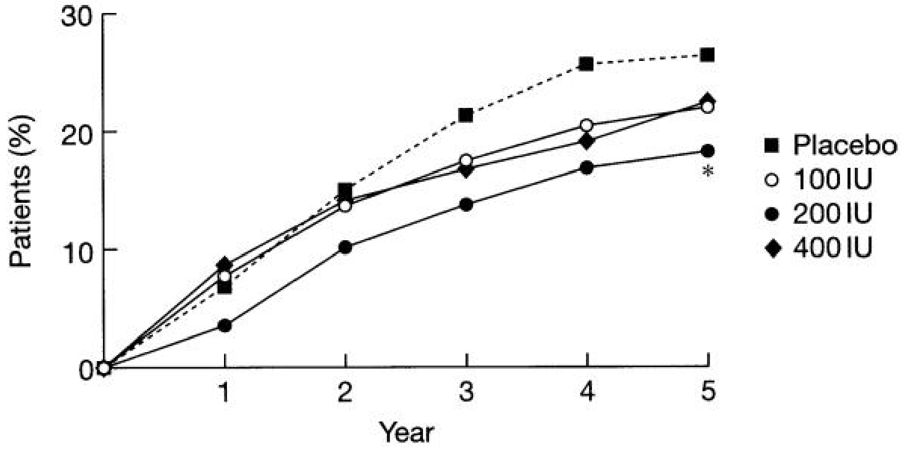
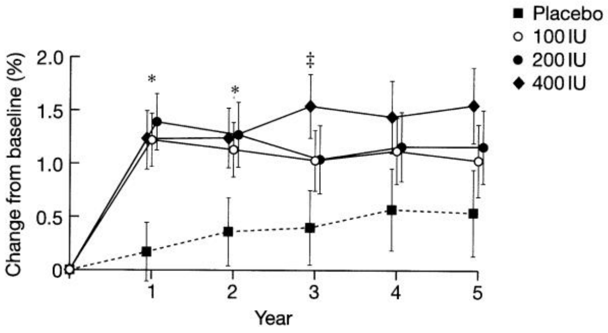


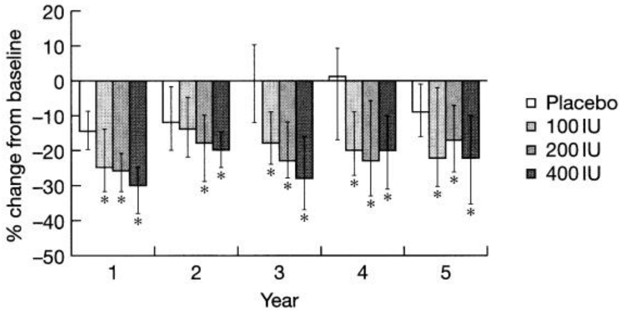
Figure 1. Cumulative percentage of participants with at least one new fracture per year. Number of participants with follow-up radiographs (placebo = 270, 100 IU = 273, 200 IU = 287, 400 IU = 278). The asterisk indicates P <0.05 versus placebo.

33% 5年随访显示 200 IU 鲑降钙素鼻喷雾剂使患者新发椎体骨折风险降低33%（RR=0.67，P=0.03）

40% 每1000例受试者每年新发椎体骨折数较安慰剂组降低40%（P=0.02）



5年内各时间点鲑降钙素组
腰椎骨密度均较基线上升（P<0.01）



200 IU组各时点CTX（骨吸收标志物）
均较安慰剂组更低（P<0.01）

Chesnut III C H, Silverman S, Andriano K, et al. A randomized trial of nasal spray salmon calcitonin in postmenopausal women with established osteoporosis: the prevent recurrence of osteoporotic fractures study[J]. The American journal of medicine, 2000, 109(4): 267-276.（Q1，IF=5.3）

QUEST试验：200 IU鲑降钙素有效稳定骨小梁微结构

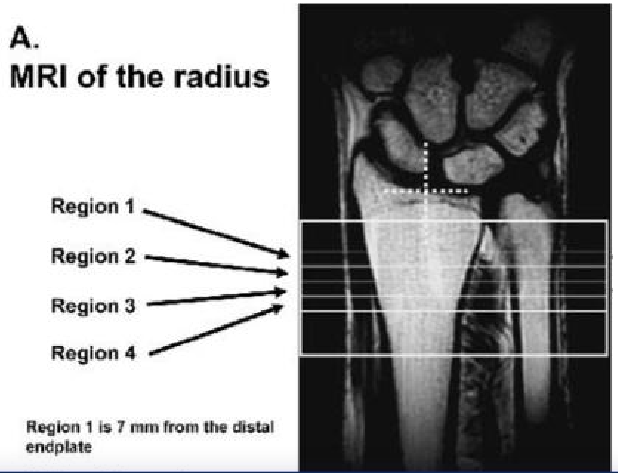
海鲸药业 Haijing Pharmaceutical

QUEST试验 2年+双盲+随机+安慰剂对照

研究设计 总计91例 → 骨质疏松症绝经后椎体骨折女性 → 安慰剂组 / 200 IU 组

所有参与者每日服用一片500 mg碳酸钙片剂

主要终点 MRI测定的骨小梁微结构参数



MRI扫描：自桡骨远端关节面起始
每个区域间隔2.5 mm

Table 4

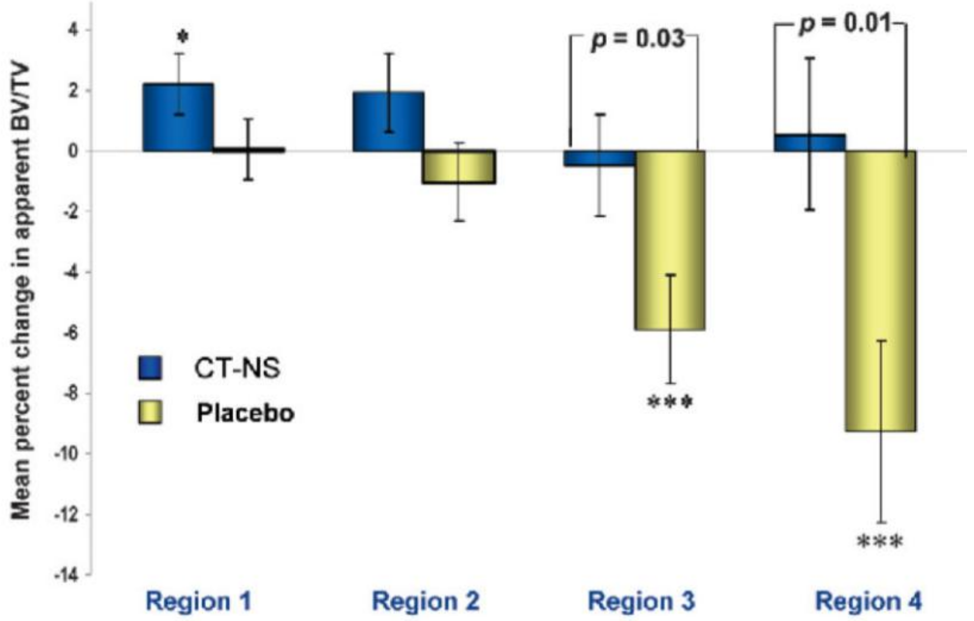
MRI Acquired Measurements of Trabecular Microarchitecture, Percentage Change at 2 Years, as a Function of BMD Change at Lumbar Spine

	Loss of BMD lumbar spine (n = 25)							Gain of BMD lumbar spine (n = 27)						
	Placebo (n = 11)			CT-NS (n = 14)			Between-group p	Placebo (n = 11)			CT-NS (n = 16)			Between-group p
	Mean	SD	Within-group p	Mean	SD	Within-group p		Mean	SD	Within-group p	Mean	SD	Within-group p	
Distal radius, region 4														
Bone V./total V. app.	–14.9	17.5	0.018	–0.5	16.2	0.901	0.045	–3.3	10.8	0.341	1.3	10.0	0.615	0.272
Trabecular number app.	–10.7	15.1	0.040	1.4	12.9	0.696	0.045	–3.0	8.4	0.260	1.9	7.6	0.335	0.126
Trabecular spacing app.	20.1	26.3	0.030	0.3	16.7	0.948	0.031	5.7	13.8	0.202	–1.4	11.9	0.638	0.165
Trabecular thickness app.	–5.1	4.8	0.006	–2.1	7.5	0.303	0.268	–0.5	4.3	0.692	–0.6	5.6	0.683	0.977
T2* decay time of the hip														
Femoral neck	9.8	14.3	0.012	–0.1	6.9	0.974	0.026	3.1	11.3	0.300	3.9	10.9	0.169	0.840
Ward's triangle	10.2	23.5	0.093	4.6	14.2	0.242	0.445	12.8	21.2	0.035	5.9	20.1	0.263	0.358
Upper trochanter	1.5	11.8	0.622	–3.2	9.0	0.211	0.240	3.7	12.6	0.268	–3.3	8.3	0.133	0.074
Lower trochanter	7.6	12.0	0.019	–0.1	8.4	0.954	0.051	10.2	8.3	0.000	–0.1	9.5	0.968	0.004

安慰剂组 安慰剂组随访期间腰椎、髋部或桡骨远端骨密度下降，伴随着骨小梁微结构的恶化

鲑降钙素组 鲑降钙素组无论骨密度上升还是下降，其骨小梁结构均保持稳定

Mean percentage change in apparent bone volume (BV/TV) baseline to 24 months



BV/TV 随访2年时，鲑降钙素组区域3、区域4表观骨体积/总体积（BV/TV）较基线变化幅度显著低于安慰剂组（P<0.05）

骨小梁 与安慰剂组相比，鲑降钙素组桡骨远端和髋关节大转子下部骨小梁微结构得到保留，而安慰剂组则出现显著恶化

Chesnut III C H, Majumdar S, Newitt D C, et al. Effects of salmon calcitonin on trabecular microarchitecture as determined by magnetic resonance imaging: results from the QUEST study[J]. Journal of bone and mineral research, 2005, 20(9): 1548-1561.（Q1，IF=6.5）

Meta分析显示，鲑降钙素具有降低血钙、改善碱性磷酸酶、PTH等作用

海鲸药业
Haijing Pharmaceutical

Li et al. European Journal of Medical Research (2021) 26:140
https://doi.org/10.1186/s40001-021-00610-x

European Journal of Medical Research

REVIEW Open Access

A meta-analysis of the therapeutic effect of intranasal salmon calcitonin on osteoporosis

浙江康复医疗中心：李宁，等

标题 鼻内鲑鱼降钙素治疗骨质疏松症疗效的Meta分析

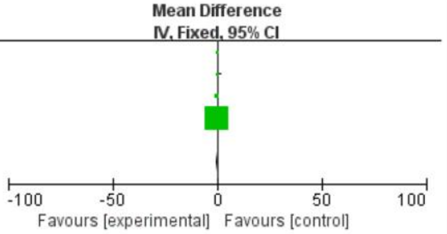
纳入文献 12篇

总样本量 1068例

结论 鼻内鲑鱼降钙素在降低血钙、提高碱性磷酸酶、降低甲状旁腺激素等方面优于常规治疗策略

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Fixed, 95% CI
Demet Ofluoglu 2006	-2.5	0.6	39	-2.8	0.9	39	3.1%	0.30 [-0.04, 0.64]
Hasan ATBİNİCİ2015	-2.4	1.1	15	-2.85	1.82	20	0.4%	0.45 [-0.52, 1.42]
R. Rizzoli A2015	-0.7	0.7	41	-0.7	0.5	42	5.1%	0.00 [-0.26, 0.26]
马荣连2009	0.87	0.11	23	0.9	0.098	20	91.4%	-0.03 [-0.09, 0.03]
Total (95% CI)			**118**			**121**	**100.0%**	**-0.02 [-0.08, 0.04]**

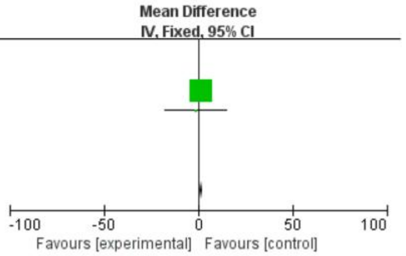
Heterogeneity: Chi² = 4.42, df = 3 (P = 0.22); I² = 32%
Test for overall effect: Z = 0.55 (P = 0.59)



鲑降钙素组对腰椎（P=0.59）、髋部骨密度（P=0.22）的影响与安慰剂组相当

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Fixed, 95% CI
Evangelia Kaskani Æ G 2005	22.23	7.79	57	22.85	9.28	45	0.0%	-0.07 [-0.46, 0.32]
Hasan ATBİNİCİ2015	96	44	15	91.5	26.7	20	0.0%	0.13 [-0.54, 0.80]
J.Y. REGINSTER 1994	85.8	3.2	142	84.2	2.7	145	99.8%	1.60 [0.91, 2.29]
Kirsten Overgaard 1989	95.45	34.47	50	96.65	47.58	50	0.2%	-1.20 [-17.49, 15.09]
M Cecchettin 1995	173	31	20	167	20	20	0.0%	0.23 [-0.40, 0.85]
R. Rizzoli A2015	-10.6	3.4	45	-12.5	3.3	45	0.0%	0.56 [0.14, 0.98]
Total (95% CI)			**192**			**195**	**100.0%**	**1.60 [0.91, 2.28]**

Heterogeneity: Chi² = 0.11, df = 1 (P = 0.74); I² = 0%
Test for overall effect: Z = 4.56 (P < 0.00001)

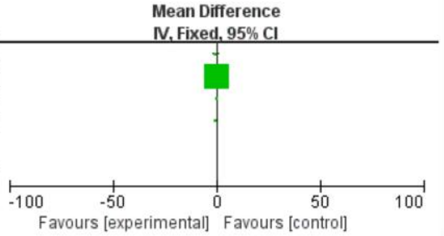


鲑降钙素组提高碱性磷酸酶浓度的效果优于常规治疗组（P<0.0001）

碱性磷酸酶：升高反映成骨细胞活跃

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Mean Difference IV, Fixed, 95% CI
Hasan ATBİNİCİ2015	2.5	1.5	15	2.8	1.8	20	0.0%	-0.30 [-1.39, 0.79]
J.Y. REGINSTER 1994	2.39	0.01	142	2.41	0.02	145	99.9%	-0.02 [-0.02, -0.02]
Kirsten Overgaard 1989	9.52	0.47	50	9.48	0.71	50	0.0%	0.04 [-0.20, 0.28]
M Cecchettin 1995	8.9	0.3	20	9.1	0.4	20	0.0%	-0.20 [-0.42, 0.02]
Total (95% CI)			**227**			**235**	**100.0%**	**-0.02 [-0.02, -0.02]**

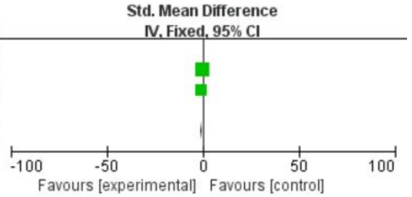
Heterogeneity: Chi² = 3.09, df = 3 (P = 0.38); I² = 3%
Test for overall effect: Z = 10.77 (P < 0.00001)



鲑降钙素组降低血钙浓度的效果优于常规治疗组（P<0.0001）

Study or Subgroup	Experimental Mean	SD	Total	Control Mean	SD	Total	Weight	Std. Mean Difference IV, Fixed, 95% CI
Apurva K.2004	0	0	0	0	0	0	0.0%	0.09 [-0.38, 0.56]
Evangelia Kaskani Æ G 2005	42.82	15.86	57	41.61	16.73	45	55.8%	0.07 [-0.32, 0.46]
R. Rizzoli A2015	3.8	0.2	45	4	0.2	45	44.2%	-0.99 [-1.43, -0.55]
Total (95% CI)			**102**			**90**	**100.0%**	**-0.40 [-0.69, -0.11]**

Heterogeneity: Chi² = 12.62, df = 1 (P = 0.0004); I² = 92%
Test for overall effect: Z = 2.67 (P = 0.008)



鲑降钙素组降低甲状旁腺激素的效果优于常规治疗组（P=0.008）

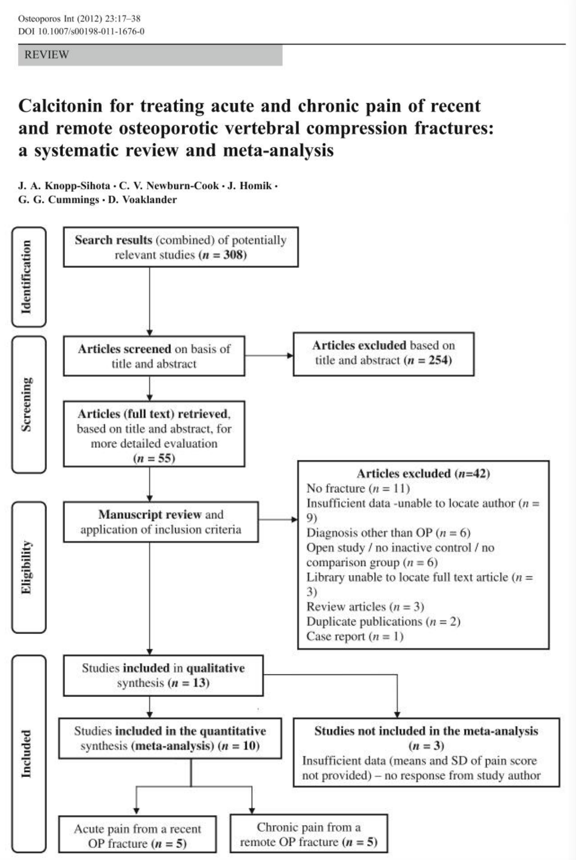
甲状旁腺激素：长期高水平PTH可能加速骨吸收

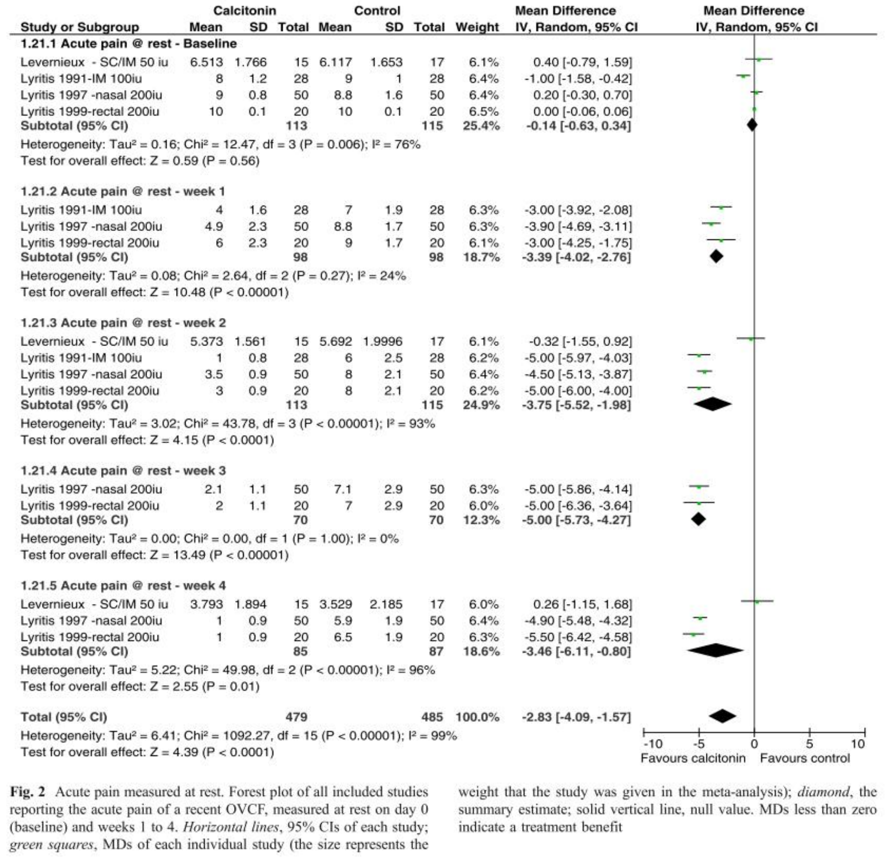
Li N, Gong Y C, Chen J. A meta-analysis of the therapeutic effect of intranasal salmon calcitonin on osteoporosis[J]. European Journal of Medical Research, 2021, 26(1): 140.（Q2，IF=4.8）

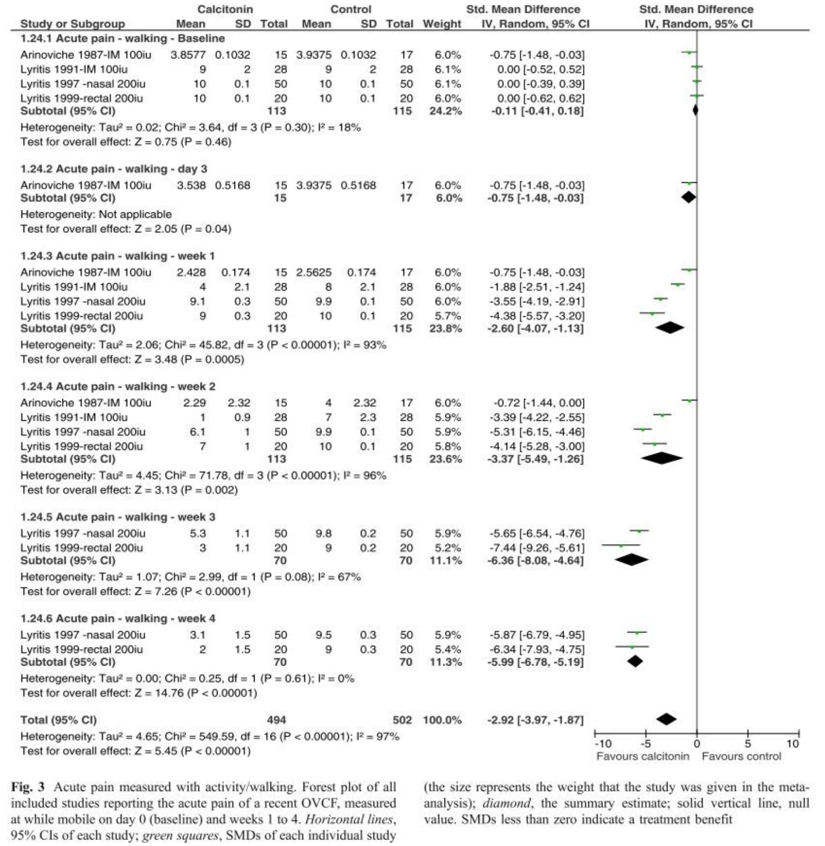
2012年Meta分析：鲑降钙素在1周内迅速缓解OVCF相关急性疼痛和慢性活动疼痛

海鲸药业
Haijing Pharmaceutical

OVCF：骨质疏松性椎体压缩性骨折







文献数 13

样本量 589

急性疼痛

- 静息痛在第1周即有所减轻[均数差（MD）=-3.39，95%CI=-4.02~-2.76]，并在4周内持续改善。
- 在第4周，活动状态下疼痛评分的差异更为显著（SMD=-5.99，95%CI=-6.78~-5.19）。

慢性疼痛

- 对于慢性疼痛患者，静息状态下各组间无统计学差异。
- 6个月时，活动状态下各组间存在较小的统计学显著差异（SMD=0.49，95%CI=-0.85至-0.13，p=0.008）。
- 副作用轻微，最常见的是肠道不适和潮红。

Knopp-Sihota J A, Newburn-Cook C V, Homik J, et al. Calcitonin for treating acute and chronic pain of recent and remote osteoporotic vertebral compression fractures: a systematic review and meta-analysis[J]. Osteoporosis International, 2012, 23(1): 17-38.（Q1，IF=5.4）

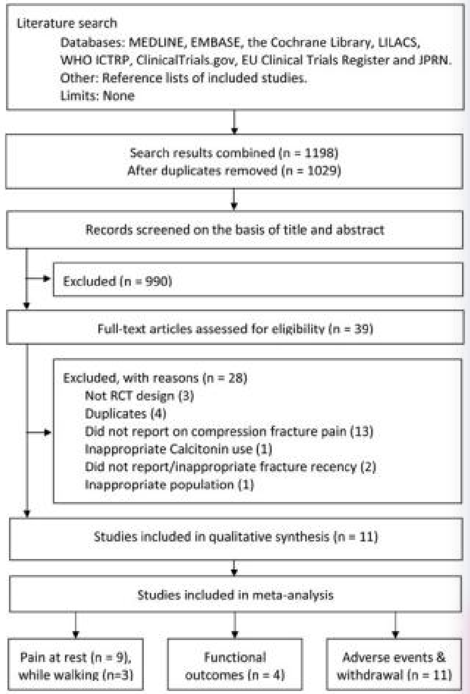
2020年Meta分析再次印证：鲑降钙素在1周内迅速缓解OVCF相关急性疼痛

海鲸药业 Haijing Pharmaceutical

OVCF：骨质疏松性椎体压缩性骨折

ORIGINAL RESEARCH • RECHERCHE ORIGINALE

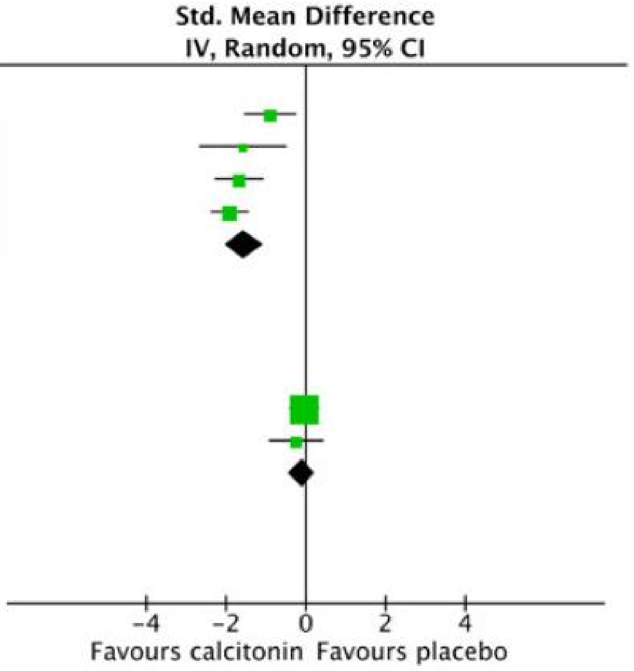
Efficacy of calcitonin for treating acute pain associated with osteoporotic vertebral compression fracture: an updated systematic review



文献数 11

样本量 713

Study or Subgroup	Favours calcitonin Mean	SD	Total	Placebo Mean	SD	Total	Weight	Std. Mean Difference IV, Random, 95% CI
1.8.1 Salmon calcitonin (100-200 IU/day)								
Lauro 1993	9.7	2.3	20	11.8	2.3	20	25.7%	-0.9 [-1.6, -0.2]
Pun 1989	1.9	0.5	9	3.0	0.9	9	13.7%	-1.6 [-2.7, -0.5]
Lyritis 1991	3.9	1.7	28	7.0	1.9	28	27.2%	-1.7 [-2.3, -1.1]
Lyritis 1997	4.9	2.3	50	8.8	1.7	50	33.3%	-1.9 [-2.4, -1.4]
Subtotal (95% CI)			**107**			**107**	**100.0%**	**-1.5 [-2.0, -1.1]**
Heterogeneity: $Tau^2 = 0.12$; $Chi^2 = 6.19$, df = 3 (P = 0.10); $I^2 = 52\%$								
Test for overall effect: Z = 6.35 (P < 0.00001)								
1.8.2 Eel calcitonin (20 IU/week)								
Tanaka 2017a	77.9		53	78.3	9.3	54	76.6%	-0.04 [-0.4, 0.3]
Tanaka 2017b	68.1	26.5	16	74.7	26.7	17	23.4%	-0.2 [-0.9, 0.4]
Subtotal (95% CI)			**69**			**71**	**100.0%**	**-0.09 [-0.4, 0.2]**
Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 0.26$, df = 1 (P = 0.61); $I^2 = 0\%$								
Test for overall effect: Z = 0.50 (P = 0.61)								



Test for subgroup differences: $Chi^2 = 24.17$, df = 1 (P < 0.00001), $I^2 = 95.9\%$

所有研究均显示，鲑降钙素能够显著减轻近期压缩性骨折相关的急性疼痛

鲑降钙素在1周时改善了疼痛评分（SMD=-1.54；95%CI=-2.02~-1.06）

依降钙素在第1周时未见改善（SMD=-0.09；95%CI=-0.42~0.25）

「降钙素不会增加不良事件（包括恶心和呕吐）的总体风险（风险比，2.10；95%CI，0.87-5.08）」

Boucher E, Rosgen B, Lang E. Efficacy of calcitonin for treating acute pain associated with osteoporotic vertebral compression fracture: an updated systematic review[J]. Canadian Journal of Emergency Medicine, 2020, 22(3): 359-367.（Q2, IF=1.7）

给药方式：鼻内给药的降钙素与肠外给药的镇痛效果相似，且耐受性更好

海鲸药业
Haijing Pharmaceutical

Analgesic Efficacy of Calcitonin for Vertebral Fracture Pain

Linsey A Blau and James D Hoehns

OBJECTIVE: To evaluate the analgesic efficacy of calcitonin for treating the pain of vertebral fractures associated with osteoporosis.

DATA SOURCES: Searches of MEDLINE (1966–July 2002), Cochrane Library, *International Pharmaceutical Abstracts* (1977–July 2002), and an extensive manual review of journals were performed using the key search terms calcitonin, analgesic, osteoporosis, vertebral fracture, and pain.

STUDY SELECTION AND DATA EXTRACTION: All articles identified from the data sources were evaluated and all information deemed relevant was included for this review.

DATA SYNTHESIS: Fractures, especially vertebral fractures, are a common complication of osteoporosis, leading to significant pain. Calcitonin has been studied for its analgesic properties. Fourteen double-blind, placebo-controlled trials that evaluated the analgesic efficacy of calcitonin for osteoporosis-related vertebral fracture pain were identified and reviewed. Thirteen of these studies demonstrated statistically significant improvement in pain or function in calcitonin-treated patients.

CONCLUSIONS: Calcitonin has proven efficacy in acute pain associated with osteoporosis-related vertebral fractures. Analgesic effects are seen with intranasal, parenteral, and rectal administration. Future studies comparing calcitonin with other commonly used analgesics are needed to more clearly define its place in therapy.

KEY WORDS: calcitonin, osteoporosis, vertebral fracture.

Ann Pharmacother 2003;37:564-70.

Published Online, 20 Feb 2003, *www.theannals.com*, DOI 10.1345/aph.1C350

降钙素在治疗骨质疏松症相关椎体骨折引起的急性疼痛方面具有确切的疗效。通过鼻内、肠外和直肠给药均可观察到镇痛效果。

Blau L A, Hoehns J D. Analgesic efficacy of calcitonin for vertebral fracture pain[J]. Annals of Pharmacotherapy, 2003, 37(4): 564-570.（Q3，IF=2.5）

鼻内给药镇痛作用起效更快：绝经后骨质疏松症研究

海鲸药业
Haijing Pharmaceutical

研究设计 双盲+双安慰剂+随机对照试验

24例疼痛性骨质疏松症绝经后女性

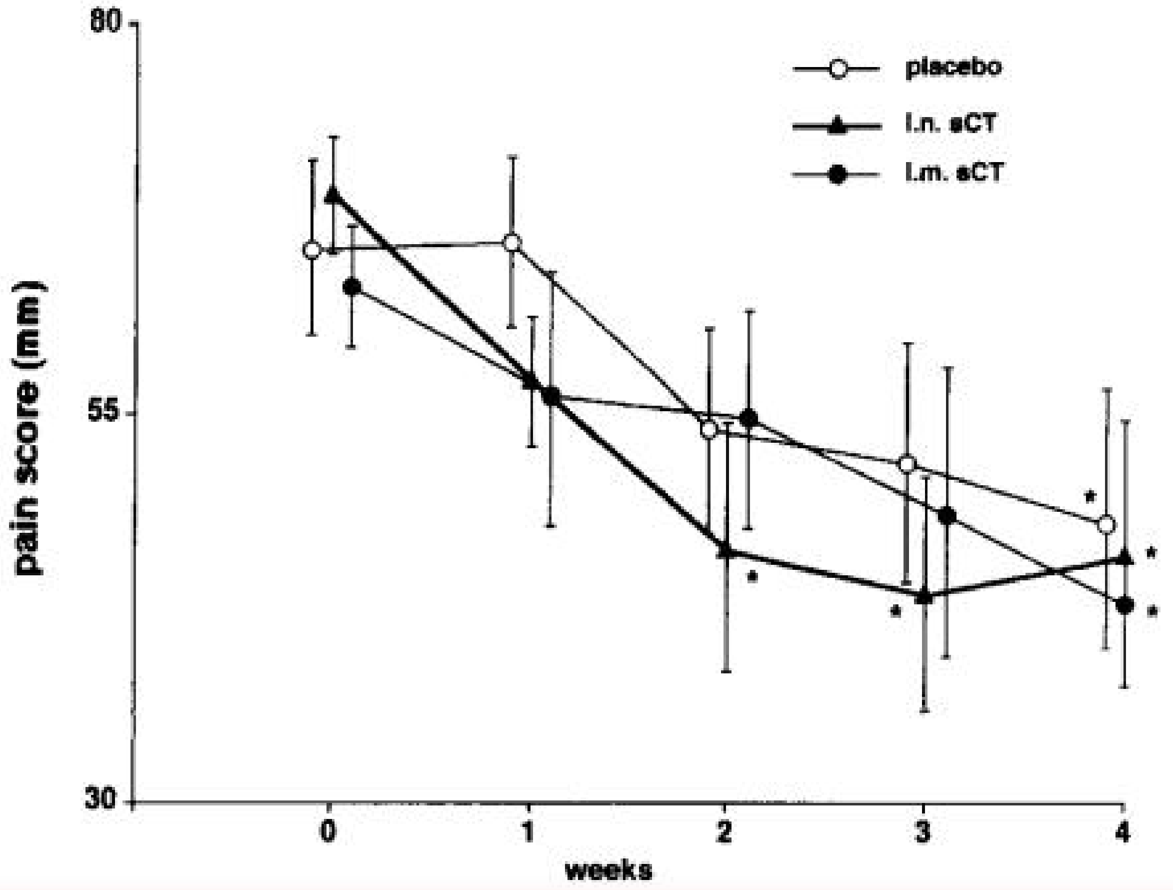
- 鼻内200 IU/d+肌注安慰剂
- 肌注100 IU/d+鼻内安慰剂
- 鼻内安慰剂+肌注安慰剂

患者每日保持
800 mg/d钙+1200 mg/d磷饮食

Analgesic effect of intranasal and intramuscular salmon calcitonin in post-menopausal osteoporosis: A double-blind, double-placebo study

A.E. Pontiroli[1], E. Pajetta[1], L. Scaglia[1], A. Rubinacci[1], G. Resmini[1], M. Arrigoni[2], and G. Pozza[1]
[1]Istituto Scientifico San Raffaele, Clinica Medica and Clinica Ortopedica, Università di Milano, Milano, [2]Istituto Ricerche LPB, Cinisello Balsamo, Milano, Italy

我们得出结论，鼻内给药可能（镇痛作用）起效更快



14 d 鼻内组（i.n.）疼痛评分自14 d起较基线显著下降，肌内组28 d显著下降

Pontiroli A E, Pajetta E, Scaglia L, et al. Analgesic effect of intranasal and intramuscular salmon calcitonin in post-menopausal osteoporosis: a double-blind, double-placebo study[J]. Aging Clinical and Experimental Research, 1994, 6(6): 459-463.（Q2，IF=4.7）

绝经后女性的雌激素替代疗法（ERT）：鲑降钙素对BMD的影响与ERT相当

海鲸药业
Haijing Pharmaceutical

REVIEW

Salmon Calcitonin Nasal Spray

An Effective Alternative to Estrogen Therapy in Select Postmenopausal Women

Louis V. Avioli

Division of Bone and Mineral Diseases, Washington University School of Medicine, at the Jewish Hospital of St. Louis, St. Louis, MO

The efficacy and safety of estrogen replacement therapy (ERT) and salmon calcitonin in the treatment of postmenopausal osteoporosis are reviewed with special consideration given to patients for whom ERT, the primary antiresorptive therapy for osteoporosis, is not indicated, tolerable, or is refused. The various formulations of estrogen and salmon calcitonin, for which the nasal spray formulation was recently approved for use in the United States, are reviewed in depth with reference to dose ranges, side effects, and convenience. Data regarding increases in bone mineral density (BMD) produced by each agent are presented. Specifically, the range of increases in BMD induced by ERT and salmon calcitonin are comparable. Given the substantial public health consequences of postmenopausal osteoporosis and osteoporotic fractures, the primary care physician is increasingly faced with the need to educate and recruit postmenopausal patients to appropriate therapy with the optimal agent for that particular patient. In the many patients who are unable or unwilling to accept, initiate, and comply with prescribed ERT, alternative therapeutic options are necessary Based on the established safety profile of salmon calcitonin, ease of administration, an uncomplicated dosing regimen, no reported drug interactions, and the lack of uterine bleeding associated with ERT or gastrointestinal adverse effects of other agents used to treat osteoporosis, salmon calcitonin nasal spray is an appropriate alternative approach for the treatment of postmenopausal bone loss.

Key Words: Estrogen replacement therapy; salmon calcitonin; postmenopausal osteoporosis.

Received June 27, 1996; Accepted July 5, 1996.

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Introduction

The likelihood that today's primary care physician will evaluate and treat female patients for symptoms of estrogen deficiency is increasing secondary to the aging of the U.S. population. In fact, it is estimated that because of increased patient longevity, the average woman will be postmenopausal for approximately one third of her life (Davidson, 1995). Although many of these female patients will present with the objective and subjective symptoms most commonly associated with menopause (e.g., hot flashes, vaginal atrophy, emotional lability), many others will present initially with a vertebral crush fracture or a Colles' fracture, indicating a more potentially devastating condition, postmenopausal osteoporosis.

Defined by the World Health Organization as "a disease characterized by low bone mass and microarchitectural deterioration of bone tissue, leading to enhanced bone fragility and a consequent increase in fracture risk," osteoporosis results from a heterogeneity of skeletal and extraskeletal factors that each contribute to the disorder (World Health Organization, 1994). Specific decreases from normal values for bone mineral density (BMD) further distinguish women with osteoporosis from those with osteopenia, or low bone mass: Osteoporosis is categorized by a BMD value 2.5 standard deviations (SD) or more below the mean for young healthy adult women, whereas osteopenia is classified by a BMD value between 1 and 2.5 SD below the reference mean; by comparison, severe (established) osteoporosis is categorized by a BMD value more than 2.5 SD below the reference mean in the presence of one or more fragility fractures (Kanis et al., 1994; World Health Organization, 1994).

The prevalence of these disorders underscores the magnitude of the public health concern related to osteoporosis within the United States today. In a 1995 report by Looker et al. (1995), in which the previously cited WHO diagnostic criteria were utilized, the estimated number of non-Hispanic white women aged 50 yr or older with

具体而言，ERT和鲑降钙素引起的BMD增加范围相当

Table 1
Prospective Studies Demonstrating Effects of Oral Estrogens on Bone Density

Author	Year	Therapy/ daily dose	No. of patients	Length of follow-up (yr)	Menopausal Age (yr)	% Change in BMD	Site measured	Method of assessment	Statistical significance
Lindsay et al.	1976	Mestranol 24 mg	63	5	0	0	Metacarpal	SPA	NA
				5	3	–0.8	Metacarpal	SPA	$p < 0.001$
				5	> 6	–0.8	Metacarpal	SPA	$p < 0.004$
Nachtigall et al.	1979	Premarin 2.5 mg	30	10	< 3	1	Metacarpal	SPA	$p < 0.001$
			37	10	> 3	0	Metacarpal	SPA	$p < 0.001$
Christiansen et al.	1980	Trisequens Forte	56	2	0.5–3.0	1.2	Distal forearm	SPA	$p < 0.01$
Lindsay et al.	1984	Premarin 0.625 mg 1.25 mg	150	2	1.5	0	Metacarpal	SPA	NS vs baseline; placebo loss $p < 0.01$ vs baseline
Ettinger et al.	1987	Premarin 0.3 mg	15	2	1.0–2.0	1.2	Lumbar spine	QCT	NA
Civitelli et al.	1988	Premarin 1.25 mg	10	1	3.0–7.0	8.3	Lumbar spine	DPA	$p < 0.05$
						2.6	Femur	DPA	$p < 0.05$
Munk-Jensen et al.	1988	Oestradiol 2 mg	50	1.5	0.5–2.0	6.4	Lumbar spine	DPA	$p < 0.01$
Gallagher et al.	1989	Premarin 0.3 mg	21	2	1.5–6.0	0.5	Lumbar spine	DPA	NA
		0.6 mg	21	2	1.5–6.0	1			
Lindsay et al.	1990	Premarin 0.625 mg	40	2	15	5.3	Lumbar spine	DPA	$p < 0.01$
						2.7	Femoral neck	DPA	NS
Stevenson et al.	1990	Premarin 0.625 mg	33	1.5	0–7.0	2	Lumbar spine	DPA	$p < 0.001$
						1.3	Femoral trochanter	DPA	$p < 0.001$
Harris et al.	1991	Estrone sulfate 0.625 mg 1.25 mg	120	2	2.5	0	Lumbar spine	QCT	$p < 0.05$ vs placebo at 18 mo
Aloia et al.	1994	Premarin 0.625 mg	118	3	0.5–6.0	1.32	Trochanter	SPA	$p < 0.005$ vs baseline
						0.1	Femoral neck	SPA	NS
						–0.23	Spine	SPA	NS
						0.54	Radius	SPA	NS
Lafferty et al.	1994	Premarin 0.625 mg	81	11.5	4.7	12	Radius (distal third)	SPA	$p < 0.02$ vs control
PEPI Trial	1996	Premarin 0.625 mg	875	3	5.0	5	Lumbar spine	DEXA	$p < 0.05$ vs baseline
						2	Femoral neck		$p < 0.05$ vs baseline

Abbreviations used: BMD = bone mineral density; SPA = single photon absorptiometry; NA = not available; NS = not statistically significant; QCT = quantitative computed tomography; DPA = dual photon absorptiometry; DEXA = dual energy x-ray absorptiometry.

Table 2
Prospective Studies Demonstrating Effects of Percutaneous Estrogens on Bone Density

Author	Year	Therapy/ daily dose	No. of patients	Length of follow-up (yr)	Menopausal Age (yr)	% Change in BMD	Site measured	Method of assessment	Statistical significance
Riis et al.	1987	Cream 3 mg	20	2	0.5–3.0	4.5	Lumbar spine	DPA	NA
Hart et al.	1987	Implant 50 mg 6/12	35	2	0–2.0	4.3	Lumbar spine	DPA	$p < 0.001$
Ribot et al.	1987	Patch 0.05 mg	30	1.5	2.9 ± 1.8 (mean)	5.4	Lumbar spine	DPA	$p < 0.02$ vs baseline
Ribot et al.	1988	Patch 50 mcg	15	2	1.0–5.0	6	Lumbar spine	DPA	$p < 0.001$ vs baseline
Adami et al.	1989	Patch 50 mcg	17	1.5	2.0–4.0	4.3	Forearm	DPA	$p < 0.01$ vs baseline
Stevenson et al.	1990	Patch 50 mcg	33	1.5	0.5–7.0	2	Lumbar spine	DPA	$p < 0.001$
						0	Femoral trochanter	DPA	$p < 0.001$ vs controls
Motta et al.	1991	Patch 50 mcg	38	2	0–14.0	0	Lumbar spine	QCT	NS loss in treated group; significant loss in control group ($p < 0.001$ vs baseline)
Lufkin et al.	1992	Patch 100 mcg	75	1	5.0–25.0	5.3	Lumbar spine	DPA	$p < 0.007$
						7.6	Femoral trochanter	DPA	$p < 0.03$
						1	Mid-radius	DPA	$p < 0.001$
Cantatore et al.	1995	Patch 0.05 mg (1st 6 mo)	12	1	< 0.5	Cortical 1.5 Trabecular 2.2	Distal forearm	SPA	Cortical NS Trabecular NS
		0.025 mg (2nd 6 mo)				Cortical –2.4 Trabecular –3.6	Distal forearm	SPA	Cortical $p < 0.03$ Trabecular $p < 0.03$
		0.1 mg (1st 6 mo)	12	1	< 0.5	Cortical 1.5 Trabecular 2.1	Distal forearm	SPA	Cortical NS Trabecular $p < 0.05$
		0.05 mg (2nd 6 mo)				Cortical 0.2 Trabecular 0.03	Distal forearm	SPA	Cortical NS Trabecular NS

Abbreviations used: BMD = bone mineral density; DPA = dual photon absorptiometry; NA = not available; QCT = quantitative computed tomography; NS = not statistically significant; SPA = single photon absorptiometry.

Table 4
Prospective Studies Demonstrating the Effects of Salmon Calcitonin Nasal Spray on Bone Density

Author	Year	Therapy/ daily dose	No. of patients	Length of follow-up (yr)	Menopausal Age (yr)	% Change in BMD	Site measured	Method of assessment	Statistical significance
Reginster et al.	1987	IN 50 IU (× 5 d/wk)	30	1	<3	0.8	Lumbar spine	DPA	NS vs baseline; $p < 0.01$ vs control
Overgaard et al.	1989	IN 100 IU	19	2	2.5–5.0	2.5	Lumbar spine	DPA	$p < 0.001$ vs placebo
						–2	Total skeleton	DPA	NS
						–2	Distal forearm	SPA	NS
						–2	Proximal forearm	SPA	NS
Overgaard et al.	1989	IN 200 IU	17	1	18.0	0.8	Proximal forearm	SPA	NS
						–1	Distal forearm	SPA	NS
						1.4	Lumbar spine	DPA	NS
Agnusdei et al.	1989	IN 100 IU	20	1	NA	5	Distal radius	DPA	$p < 0.01$
Overgaard et al.	1992	IN 50 IU	40	2	21	–1	Distal forearm	SPA	NS
						1	Lumbar spine	Dual energy X-ray absorptiometry	$p = 0.008$
		IN 100 IU	43	2	24	–1	Distal forearm	SPA	NS
						2	Lumbar spine	Dual energy X-ray absorptiometry	$p = 0.008$
		IN 200 IU	41	2	21	–0.9	Distal forearm	SPA	NS
						3	Lumbar spine	Dual energy X-ray absorptiometry	$p = 0.008$
Reginster et al.	1994	IN 50 IU (× 5 d/wk)	91	3	<3	1.8	Lumbar spine	DPA	$p < 0.05$ vs baseline
Ellerington et al.	1996	IN 200 IU	18	2	<5	0.5	Lumbar spine	DPA	NS vs placebo
		daily	18	2	>5	3.1	Lumbar spine	DPA	$p < 0.01$ vs placebo
		IN 200 IU	17	2	<5	–0.1	Lumbar spine	DPA	NS vs placebo
		(3×/wk)	18	2	>5	0.6	Lumbar spine	DPA	NS vs placebo

Abbreviations used: BMD = bone mineral density; IN - intranasal; DPA = dual photon absorptiometry; NS = not statistically significant; SPA = single photon absorptiometry.

Table 3
Prospective Studies Demonstrating the Effects of Injectable Salmon Calcitonin on Bone Density

Author	Year	Therapy/ daily dose	No. of patients	Length of follow-up (yr)	Menopausal Age (yr)	% Change in BMD	Site measured	Method of assessment	Statistical significance
Mazzuoli et al.	1986	IM 100 IU (qod)	21	0.5	17.3	10	Distal radius	DPA	$p < 0.001$
			16	1		13			$p < 0.05$
Civitelli et al.	1988	SC 50 IU (qod)	53	1	7.0–9.0	7.4	Lumbar spine	DPA	$p < 0.001$
						–3	Femoral mid-shaft	DPA	$p < 0.01$
Agnusdei et al.	1989	SC 100 IU (qod)	10	1	NA	7	Distal radius	DPA	$p < 0.01$
Mazzuoli et al.	1990	IM 100 IU (qod)	13	0.5	0	0	Distal forearm	DPA	NS vs baseline; $p < 0.01$ vs control
Meschia et al.	1992	IM 40 IU (× 2 d/wk)	26	1	5	0	Lumbar spine	DPA	NS
		IM 40 IU (× 2 d/wk) + Premarin 1.25 mg	26	1	4	10	Lumbar spine	DPA	$p < 0.001$
Rico et al.	1995	IM 100 IU (× 10 d/mo)	36	2.0	18	30.7	Pelvis	DPA	$p < 0.001$
						6.2	Arm	DPA	$p < 0.001$

Abbreviations used: BMD = bone mineral density; IN - intramuscular; DPA = dual photon absorptiometry; NS = not statistically significant; SC = subcutaneous; SPA = single photon absorptiometry.

Avioli L V. Salmon calcitonin nasal spray: An effective alternative to estrogen therapy in select postmenopausal women[J]. Endocrine, 1996, 5(2): 115-127.（Q3，IF=2.9）

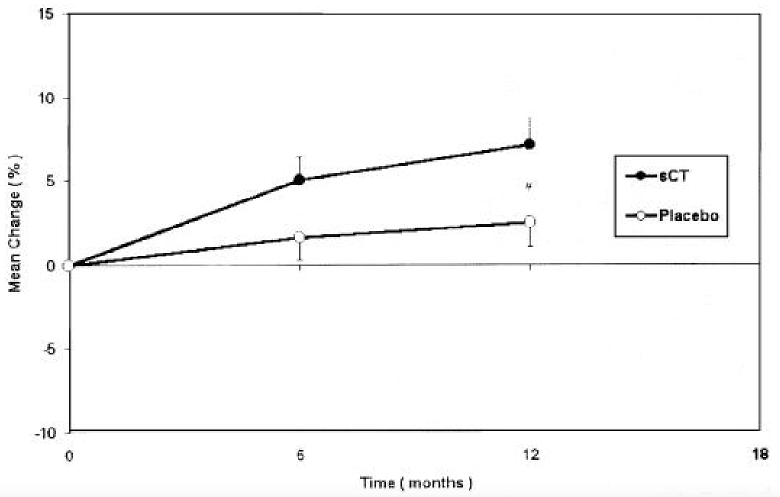
男性骨质疏松症患者同样可以获益：腰椎BMD增加、抑制骨吸收

海鲸药业
Haijing Pharmaceutical

研究设计 随机+双盲+安慰剂对照试验

28例骨质疏松症**男性**
- 每日鼻内200 IU 鲑降钙素
- 每日鼻内200 IU 安慰剂

患者每日补充500 mg钙



12个月后降钙素组的腰椎骨密度增加值（7.1%±1.7%）显著大于安慰剂组（2.4%±1.5%）（P<0.05）

TABLE 2. CHANGES IN BONE MARKERS (MEAN ± SEM)

	Baseline	Change from baseline (%) 3 Months	6 Months	9 Months	12 Months
Urinary-DPD/Cr (nmol/mmol)					
Placebo	6.05	1.3 ± 3.6	−4.0 ± 4.5	−1.0 ± 6.4	−4.8 ± 5.37
SCT (200 IU)	6.55	−24.9 ± 3.7$^{\P,\#}$	−24.5 ± 4.8$^{\S,\|}$	−29.8 ± 4.5$^{\S,\#}$	−36.9 ± 5.9$^{\S,\#}$
Urinary-NTX/Cr (nmol/BCE per mmol)					
Placebo	55.3	5.1 ± 3.2	1.3 ± 4.8	−6.0 ± 3.1	−4.1 ± 3.8
SCT (200 IU)	62.7	−17.1 ± 5.6*,†	−23.0 ± 6.5†,‡	−36.8 ± 5.1$^{\|,\P}$	−46.8 ± 7.6$^{\|,\P}$
Urinary-CTX/Cr (μg/mmol)					
Placebo	299.6	−4.7 ± 3.0	−5.2 ± 1.3	−6.1 ± 3.6	−4.4 ± 1.5
SCT (200 IU)	308.7	−19.7 ± 5.1*,†	−29.5 ± 7.7‡,§	−39.5 ± 5.5$^{\P,\#}$	−48.1 ± 7.1$^{\P,\#}$
BALP (μg/L)					
Placebo	12.9	−8.3 ± 7.1	−10.9 ± 9.3	3.7 ± 9.9	−1.1 ± 9.9
SCT (200 IU)	13.4	−4.1 ± 6.8	−18.3 ± 5.6‡	−17.7 ± 7.7‡	−17.3 ± 8.0‡
OC (ng/ml)					
Placebo	12.2	−1.9 ± 1.5	−0.2 ± 1.7	−2.3 ± 2.1	−4.6 ± 2.1
SCT (200 IU)	12.7	0.8 ± 2.2	−1.1 ± 2.2	−4.0 ± 1.9‡	−5.1 ± 1.6‡
PICP (μg/L)					
Placebo	119.9	0.2 ± 4.22	−2.8 ± 4.4	−2.6 ± 5.2	−2.8 ± 6.6
SCT (200 IU)	111.8	8.4 ± 6.1	−7.5 ± 7.6	−2.6 ± 8.9	−25.6 ± 2.97$^{\dagger,\#}$
PINP (μg/L)					
Placebo	38.6	−2.3 ± 3.9	−3.17 ± 6.9	−1.26 ± 6.8	−0.6 ± 6.8
SCT (200 IU)	47.4	−11.9 ± 4.1*	−11.8 ± 8.6*	−16.4 ± 5.4†,‡	−29.8 ± 4.7$^{\S,\#}$

* $p < 0.05$ compared with baseline; † $p < 0.01$ compared with placebo; ‡ $p < 0.01$ compared with baseline; § $p < 0.001$ compared with placebo; ‖ $p < 0.001$ compared with baseline; ¶ $p < 0.0005$ compared with placebo; # $p < 0.0005$ compared with baseline.

NTX&CTX：骨吸收标志物
NTX：I型胶原交联N-末端肽
CTX：I型胶原交联C-末端肽

鼻喷鲑降钙素治疗导致骨吸收标志物显著抑制，而对骨形成标志物的抑制作用程度较轻，而安慰剂组则未见此效应

Dr. Trovas G P, Lyritis G P, Galanos A, et al. A randomized trial of nasal spray salmon calcitonin in men with idiopathic osteoporosis: effects on bone mineral density and bone markers[J]. Journal of Bone and Mineral Research, 2002, 17(3): 521-527.（Q1，IF=6.5）

JAMA子刊Meta分析：降钙素和NSAIDs缓解短期疼痛显著有效且降钙素更优

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JAMA Network Open

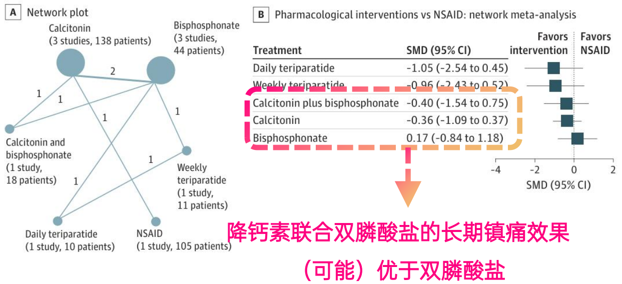
Original Investigation | Geriatrics

Conservative Treatments in the Management of Acute Painful Vertebral Compression Fractures
A Systematic Review and Network Meta-Analysis

Assil-Ramin Alimy, MD; Athanasios D. Anastasilakis, MD; John J. Carey, MB, BCh, BAO, MS; Stella D'Oronzo, MD, PhD; Anda M. Naciu, MD, PhD; Julien Paccou, MD, PhD; Maria P. Yavropoulou, MD, PhD; Willem F. Lems, MD, PhD; Tim Rolvien, MD, PhD

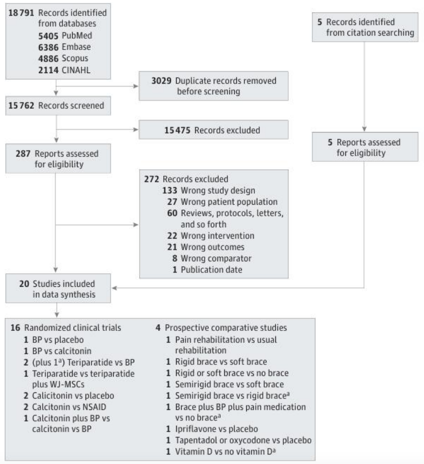
纳入 20项临床研究 | 样本量 2102例 急性椎体压缩性骨折患者

Figure 3. Pharmacological Interventions for Long-Term Pain Management (Nonspecified Pain)



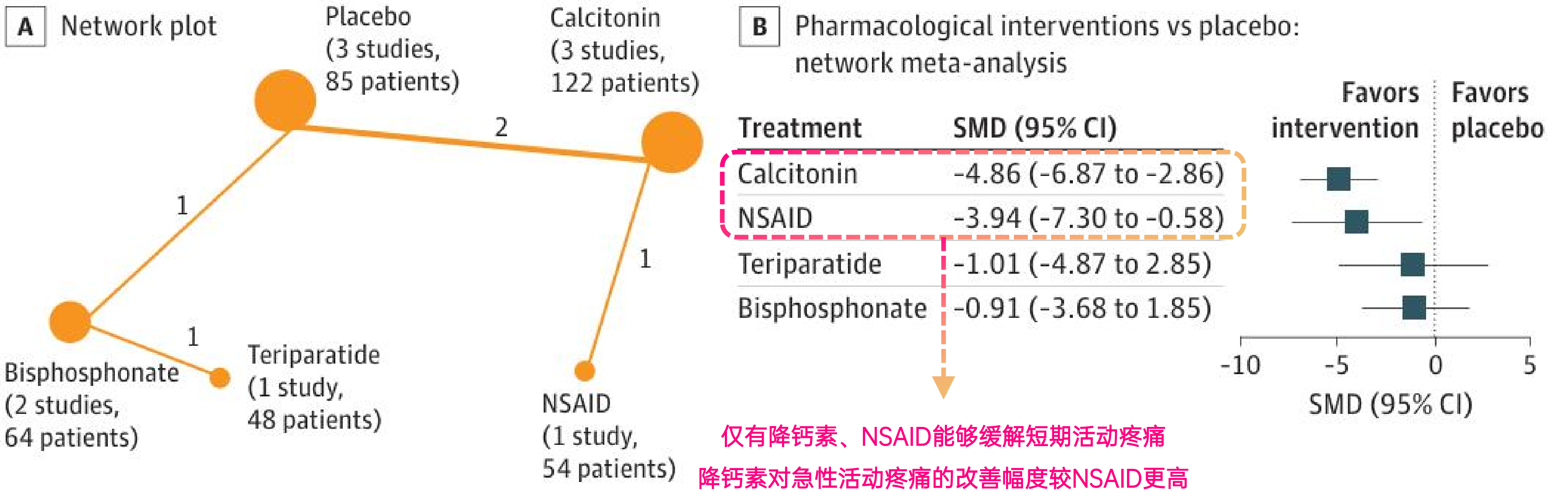
- 最近一次随访的非特异性疼痛：各药物均不具备较对照组的统计学差异
- 作者指出：「对于长期非特异性疼痛管理，与每日（SMD=1.21；95%CI 0.11~2.31）或每周（SMD=1.13；95%CI 0.05~2.21）给药的特立帕肽相比，双膦酸盐的镇痛效果较差，且没有一种治疗方法优于NSAIDs」

Figure 1. Flowchart of Included Studies



关注急性椎体压缩性骨折患者
短期（4周）活动疼痛
长期（最近1次随访）疼痛

Figure 2. Short-Term Pain During Activity



Alimy A R, Anastasilakis A D, Carey J J, et al. Conservative treatments in the management of acute painful vertebral compression fractures: a systematic review and network meta-analysis[J]. JAMA network open, 2024, 7(9): e2432041.（Q1, IF=11.7）

复习：

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鲑降钙素鼻喷雾剂

- 是什么
  - 初识降钙素 — 调控血钙水平的多肽激素
  - 降钙素的生理作用
    - 调节钙稳态 — 降低血清钙浓度
    - 作用于破骨细胞 — 减少骨吸收
    - 作用于肾小管 — 抑制钙磷重吸收
    - 作用于中枢&外周神经系统 — 改善疼痛
  - 鲑降钙素的来源 — 鲑鱼
    - 鳃后体/终末腮体 — 32个氨基酸组成的激素
    - 效价为人降钙素的40~50倍 — 原因：海水高钙
  - 降钙素的药理作用
    - 降钙
    - 抑制骨吸收
    - 镇痛
  - 降钙素的药理机制 — 降低破骨细胞活性、减少骨质流失、中枢神经机制、炎症介质机制等
  - 鲑降钙素的药代动力学
    - 吸收、分布、代谢、排泄
      - 鼻内给药生物利用度为3%~5%
      - 给药中位10 min达到血浆峰浓度
      - 无蓄积效应，主要在肾脏内代谢
    - 鼻喷剂型的特点、优势、循环过程 — 鼻腔血管丰富、降低全身副作用风险、非侵入性、便利给药
  - 鲑降钙素的安全性：恶性肿瘤风险上升? — 主要受PROOF研究影响，但无生物学合理性，指南推荐短期用药
  - 降钙素家族：鲑降钙素vs依降钙素? — 鲑降钙素缓解疼痛效果更优
- 为什么
  - 临床试验
    - PROOF试验：200 IU鲑降钙素有效预防骨质疏松性骨折复发
    - QUEST试验：200 IU鲑降钙素通过改善骨小梁微结构降低骨折风险
  - 系统评价与Meta分析
    - Meta分析：鲑降钙素具有降低血钙、改善碱性磷酸酶、PTH等作用
    - 2012年Meta分析：1周内迅速缓解OVCF相关急性疼痛和慢性活动疼痛
    - 2020年Meta分析：1周内迅速缓解OVCF相关急性疼痛
    - 鼻内给药的降钙素与肠外给药的镇痛效果相似，且耐受性更好
    - 绝经后骨质疏松症研究：鼻内给药镇痛作用起效更快
    - 绝经后女性的雌激素替代疗法（ERT）：鲑降钙素改善BMD与ERT相当
    - 男性骨质疏松症患者同样可以获益：腰椎BMD增加、抑制骨吸收
    - JAMA Network Open — 降钙素和NSAIDs缓解短期疼痛显著有效且降钙素更优

核心内容：三重药理机制的骨松相关骨痛短期鼻喷辅助用药

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是什么

- 兼具降钙+抑制骨吸收+镇痛三重药理作用的多肽激素
- 依从性和耐受性更高的鼻喷雾剂

作用机制

- 抑制肾小管钙磷重吸收、促进钙磷排泄
- 结合破骨细胞抑制骨吸收
- 中枢（提高疼痛阈值）+外周（减轻疼痛刺激）双重作用镇痛

循证证据

- 1周内迅速减轻骨折相关急性疼痛
- 抑制骨吸收/稳定骨小梁结构预防骨折复发

安全性

- 不良事件总体风险低
- 恶性肿瘤风险轻微升高无因果证据，指南推荐短疗程（3个月）使用

巩固小结：场景·机制·定位

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鲑降钙素鼻喷雾剂

短期镇痛、辅助抑制骨吸收

骨松患者综合康复管理

适用场景

① 骨质疏松相关骨痛
② 近期椎体压缩骨折急性疼痛
③ 骨科病房卧床/制动相关急性骨丢失风险

药理机制

① 抑制骨吸收：作用于破骨细胞，减少骨吸收，辅助管理急性骨丢失
② 缓解骨痛：中枢+外周双通路共同参与

建议定位

① 短期辅助，不替代长期一线抗骨质疏松治疗
② 严格遵循说明书、短疗程、有效剂量

随堂测试：关于鲑降钙素鼻喷雾剂的单项选择题

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1 降钙素抗骨吸收主要机制是什么？

A、促进甲状旁腺激素分泌　B、抑制破骨细胞
C、促进血钙肠道吸收　D、促进破骨细胞活性提升

2 降钙素能够促进肾脏对___的排泄？

A、钙、磷　B、钾
C、肌酐　D、维生素D

3 鲑降钙素的效价是人降钙素的_倍？

A、10~20　B、20~30
C、30~40　D、40~50

4 鲑降钙素对哪类疼痛的改善最为明显？

A、偏头痛　B、骨关节炎
C、椎体骨折相关疼痛　D、类风湿性关节炎

5 鲑降钙素鼻喷较肌注的最大优势在于？

A、疼痛缓解更早　B、生物利用度更高
C、侵入性更高　D、国内外指南推荐

https://pharmacyfreak.com/mechanism-of-action-of-calcitonin

随堂测试：关于鲑降钙素鼻喷雾剂的单项选择题

海鲸药业 Haijing Pharmaceutical

1. 降钙素抗骨吸收的主要机制是什么？
   - A、促进甲状旁腺激素分泌
   - B、抑制破骨细胞
   - C、促进血钙肠道吸收
   - D、促进破骨细胞活性提升

2. 降钙素能够促进肾脏对___的排泄？
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   - B、钾
   - C、肌酐
   - D、维生素D

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   - B、生物利用度更高
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   - D、国内外指南推荐

https://pharmacyfreak.com/mechanism-of-action-of-calcitonin

后续医学支持/培训计划：循证证据检索+推广方向细化

海鲸药业
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1

循证证据持续检索

基于产品定位
进一步检索文献证据

与当前抗骨吸收药物的横向比较（机制/靶点/获益/证据/适用人群）...

与当前抗骨质疏松（抗骨吸收/促骨形成/...）药物的联合使用（安全性/有效性/经济性）...

感冒、流涕患者的使用舒适度、疗效...

2

科室/病种针对性培训策略

结合当前科室/疾病痛点
制定医学策略并开展培训（注：以下内容仅供参考）

科室	人群	切入点
骨科/脊柱外科	骨质疏松性椎体骨折、骨折后卧床/急性疼痛、活动受限...	缓解骨质疏松骨折相关性骨痛，促进早期活动
创伤/骨关节科	围手术期疼痛、卧床制动伴疼痛	疼痛管理+抗骨吸收辅助治疗
骨质疏松门诊	骨质疏松伴骨痛	短期疼痛管理
内分泌科	CKD伴骨质疏松性疼痛	肾功能不全者友好+镇痛
其他	康复科、疼痛科、老年医学科，等	三重机制短期用药

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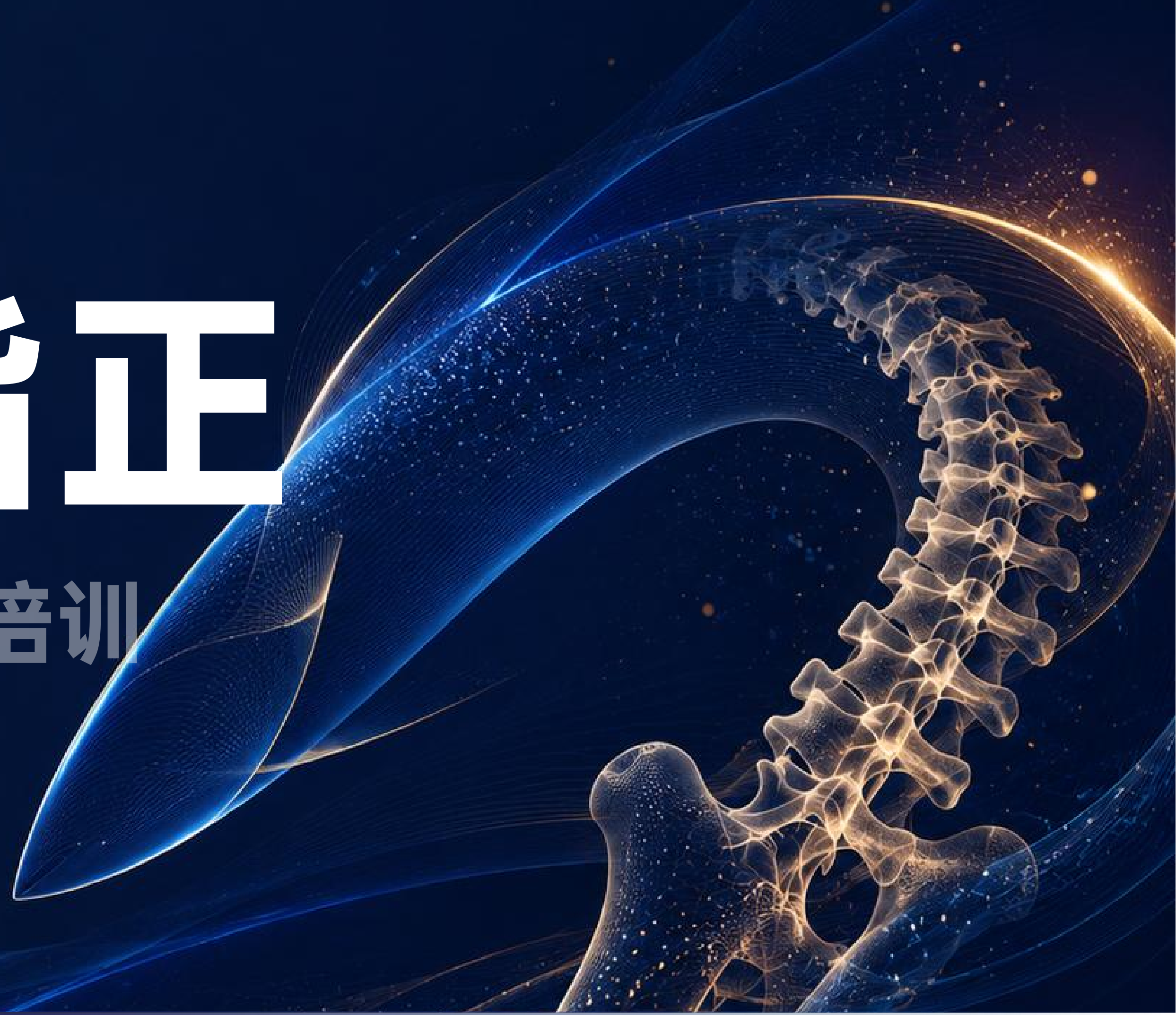
敬请批评指正

鲑降钙素鼻喷雾剂 医学知识培训



海鲸药业·中央市场医学部·医学部 王旻昊

2026-06-22